

TICVAI

Payment & Payment Orchestration

Professional Functional Design & Configuration Reference

Module 3

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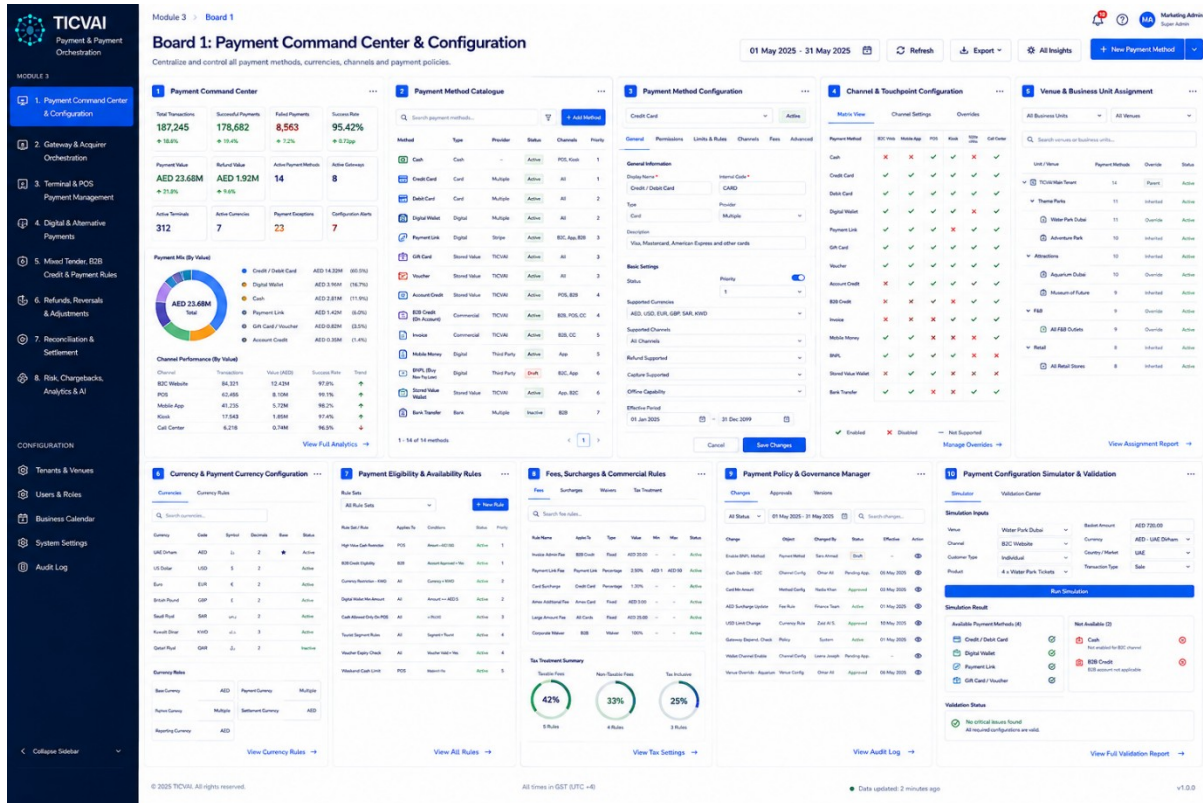
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Module 3 — Payment & Payment Orchestration

Board 1 — Payment Command Center & Configuration



Board Objective

Board 1 establishes the **central configuration and operational control layer** for all payments across TICVAI.

It defines:

Which payment methods are available, where they are available, under what conditions they can be used, and how payment configuration is governed across tenants, venues, channels, currencies and transaction types.

This board should **not** perform detailed gateway routing, terminal management, refunds, reconciliation, or fraud management. Those belong to subsequent Payment boards.

Board 1 becomes the master configuration layer consumed by:

- B2C
- POS
- Mobile POS
- Kiosk
- Mobile App
- B2B
- Call Center
- F&B
- Retail
- Membership
- Wallet
- API/Partner channels

Board 1 shall contain 10 backend screens.

Screen 1 — Payment Command Center

Purpose

Provide centralized visibility into payment operations and configuration across the tenant.

KPI Cards

- Total Payment Transactions
- Successful Payments
- Failed Payments
- Payment Success Rate
- Payment Value
- Refund Value
- Active Payment Methods
- Active Gateways
- Active Terminals
- Active Currencies
- Payment Exceptions
- Configuration Alerts

Payment Mix

Show payment volume/value by:

- Credit/Debit Card
- Cash
- Digital Wallet
- Payment Link
- Gift Card
- Voucher
- Account Credit
- B2B Credit
- Other configured tender

Channel Performance

Example:

Channel	Transaction s	Value	Success
B2C	84K	AED 12.4M	97.8%
POS	62K	AED 8.1M	99.1%
App	41K	AED 5.7M	98.2%
Kiosk	17K	AED 1.9M	97.4%

Operational Alerts

Examples:

- Gateway unavailable
- Payment method disabled
- Terminal offline
- Abnormal failure rate
- Currency configuration missing
- Settlement issue

These alerts may originate from later Payment boards but should be summarized here.

Screen 2 — Payment Method Catalogue

Purpose

Maintain the centralized catalogue of payment/tender types supported by TICVAI.

Payment Method Types

Support configuration for:

Cash

- Cash
- Petty cash where applicable

Cards

- Credit card
- Debit card

Digital Payments

- Digital wallet
- Mobile wallet
- Online payment
- Payment link

Stored / Controlled Value

- Gift card
- Voucher
- TICVAI Wallet / Stored Value
- Account credit

Commercial / B2B

- B2B credit
- Invoice
- Corporate account
- On-account payment

Other

Extensible client-specific payment methods.

Payment Method Record

Each method should contain:

- Payment Method ID
- Name

- Display Name
- Type
- Provider

- **Status**

- Currency support
- Refund capability
- Recurring capability
- Online/offline capability
- Channels
- Effective dates

Status

- Draft
- Active
- Suspended
- Expired
- Archived

Screen 3 — Payment Method Configuration

Purpose

Configure detailed behavior for each payment method.

General Configuration

- Payment method name
- Internal code
- Customer-facing label
- Description
- Icon
- Tender category
- Provider
- Effective dates

Transaction Permissions

Configure whether the method allows:

- Sale
- Refund
- Partial refund
- Void
- Reversal
- Recurring payment
- Preauthorization where supported
- Capture where supported

Amount Controls

- Minimum transaction
- Maximum transaction
- Daily limits where required
- Currency restrictions

Example

B2B Account Credit

Minimum:

AED 0

Maximum:

Based on available account credit.

Channels:

B2B Portal / POS / Call Center.

Refund:

Account credit.

Screen 4 — Channel & Touchpoint Payment Configuration

Purpose

Control which payment methods are available on each sales channel.

Channel Matrix

Payment Method	B2C POS		App	Kiosk	B2B	Call Center
Card	✓	✓	✓	✓	✓	✓
Cash	—	✓	—	✓*	—	✓*
Wallet	✓	✓	✓	✓	—	✓
Payment Link	✓	✓	✓	—	✓	✓
B2B Credit	—	✓	—	—	✓	✓

*Subject to supported hardware/operational configuration.

Configuration Dimensions

Payment availability can vary by:

- Tenant
- Venue
- Location
- Sales channel
- POS type
- Product type
- Customer type
- Transaction type

Example

Cash

Allowed:

Main Ticketing POS.

Not allowed:

B2C / Mobile App.

Screen 5 — Venue, Location & Business Unit Payment Assignment

Purpose

Allow different venues and business units to operate different payment configurations.

This is important for TICVAI's multi-venue architecture.

Hierarchy

Tenant

↓

Legal Entity / Business Unit

↓

Venue

↓

Location

↓

Sales Point

Example

Venue A

Supports:

- Cash
- Visa/Mastercard
- Digital Wallet
- Gift Card

Venue B

Supports:

- Card
- Digital Wallet

No cash.

Business Area Configuration

Allow payment methods to differ across:

- Ticketing
- F&B
- Retail
- Rental
- Membership
- Parking
- Other TICVAI modules

Inheritance

Configuration should support:

Tenant Default

↓

Venue Override

↓

Location Override

↓

Channel Override

with clear visibility into where the final configuration came from.

Screen 6 — Currency & Payment Currency Configuration

Purpose

Define which currencies can be used for payment and how payment currencies interact with TICVAI's selling and accounting currencies.

Currency Configuration

For each currency:

- ISO Currency Code
- Name
- Symbol
- Decimal precision
- **Status**
- Effective dates

Currency Roles

Distinguish:

- Base currency
- Selling currency
- Payment currency
- Settlement currency
- Reporting currency

Example

Product displayed:

USD 100

Customer payment:

AED 367.25

Gateway settlement:

AED

Reporting:

AED

Currency Rules

Configure:

- Accepted currencies by channel
- Accepted currencies by payment method
- Venue restrictions
- Gateway compatibility
- Currency conversion requirement

Important Boundary

Board 1 defines **currency/payment configuration**.

The detailed exchange-rate engine and Finance accounting treatment should remain within their authoritative modules.

Screen 7 — Payment Eligibility & Availability Rule Builder

Purpose

Determine whether a payment method should be available for a particular transaction.

Rule Inputs

Support conditions such as:

- Channel
- Venue
- Location
- Product
- Product category
- Transaction amount
- Currency
- Customer type
- Membership
- B2B account
- Country/market where applicable
- Transaction type

Example

IF

Channel = B2B

AND

Customer Account Status = Approved

AND

Available Credit $\geq$ Transaction Total

THEN

Enable:

B2B Credit

Another Example

IF

Transaction Amount > AED 10,000

THEN

Disable:

Cash

where required by tenant policy.

Logic

Support:

- AND
- OR
- NOT
- Include
- Exclude

Rule Priority

When multiple rules apply:

- Priority
- Specificity
- Effective dates
- Override policy

must determine the result.

Screen 8 — Payment Fees, Surcharges & Commercial Rules**Purpose**

Configure payment-related commercial rules where legally and contractually permitted.

Configuration

Support:

- Fixed payment fee
- Percentage fee
- Minimum fee
- Maximum fee
- Payment-method-specific fee
- Channel-specific fee
- Currency-specific fee
- Waiver rule

Example

Payment Method:

Corporate Invoice

Administration Fee:

AED 20

Channel:

B2B.

Another Example

Payment Link:

Fee:

Configured according to tenant commercial policy.

Tax Handling

Configure whether a payment fee is:

- Taxable
- Non-taxable
- Tax inclusive
- Tax exclusive

while the Tax Engine remains authoritative for actual tax calculation.

Important

The system should not assume surcharging is universally permitted.

Rules must be configurable according to the client's applicable commercial and regulatory requirements.

Screen 9 — Payment Policy, Governance & Approval Manager

Purpose

Provide governance over sensitive payment configuration changes.

Governed Objects

Include:

- Payment methods
- Channel assignments
- Currency configuration
- Payment limits
- Eligibility rules
- Payment fees
- Provider assignments
- **Critical payment settings**

Lifecycle

Draft

→ **Review**

→ **Approval**

→ **Scheduled**

→ **Active**

→ **Suspended**

→ **Archived**

Approval Example

Changing:

B2C Card Provider

may require:

Commercial Manager

-

Finance Manager

-

System Administrator.

Effective Dating

Support:

- Effective from
- Effective until
- Scheduled activation
- Scheduled expiry

Version Control

Every major configuration change should generate a new version.

Audit

Capture:

- Changed by
- Date/time
- Old value
- New value
- Reason
- Approval
- Effective date

Screen 10 — Payment Configuration Simulator & Validation Center**Purpose**

Allow administrators to test payment configuration before publishing it.

This is especially important because payment configuration errors can immediately stop sales.

Simulation Inputs

Select:

- Tenant
- Venue
- Channel
- Customer type
- Product

- Basket value
- Currency
- Country/market where relevant
- Transaction type

Example**Scenario**

Venue:

Water Park

Channel:

B2C

Customer:

Individual

Basket:

AED 720

Currency:

AED

Result

Available payment methods:

- ✓ Visa/Mastercard
- ✓ Digital Wallet
- ✓ Payment Link
- ✓ Gift Card

Unavailable:

- × Cash
- × B2B Credit

Explanation**Cash**

Not supported on B2C.

B2B Credit

Customer does not have an eligible B2B account.

Configuration Validation

Before activation, automatically check for:

Missing Configuration

- No payment method available
- No gateway mapped
- Unsupported currency
- Missing settlement configuration
- Missing refund capability

- Invalid effective dates

Conflicting Configuration

Example:

Payment Method:

Card

Channel:

B2C = Active

but:

No active gateway supports B2C/AED.

Result:

Critical Configuration Error

Cannot Publish

Payment Configuration Hierarchy

I recommend to implement a clear configuration hierarchy:

Global Platform Capability

↓

Tenant

↓

Legal Entity / Business Unit

↓

Venue

↓

Location

↓

Channel

↓

Payment Method

↓

Transaction Context

This avoids creating hundreds of duplicated payment configurations.

Inheritance & Override Model

Example:

Tenant Default

Card = Enabled.

↓

Venue Override

Venue B disables cash.

↓

Channel Override

Kiosk enables Card + Wallet only.

↓

Transaction Rule

B2B transaction enables Account Credit.

The UI should clearly show:

Effective configuration inherited from Tenant

or:

Overridden at Venue level

This will be very important for administrators.

Core Payment Decision Flow

When checkout asks:

“Which payment methods can I show?”

TICVAI should process:

Transaction Context

↓

Tenant / Venue Configuration

↓

Channel Configuration

↓

Currency Compatibility

↓

Payment Eligibility Rules

↓

Customer / B2B Eligibility

↓

Payment Method Availability

↓

Provider Availability

↓

Commercial Rules

↓

Available Payment Methods

The customer-facing channel receives only the final permitted methods.

Critical Architecture Boundary

Board 1 should **not directly implement gateway transaction processing**.

It defines:

What payment methods can be used.

Board 2 will determine:

Which gateway/acquirer should process the transaction.

Board 3 will manage:

Which physical terminal/device handles card-present payments.

This separation is important.

Payment Method vs Gateway

These must not be treated as the same thing.

Payment Method

What the customer selects:

Credit Card**Gateway**

How TICVAI processes it:

Gateway A / Gateway B / Acquirer C.

One payment method may have several gateways.

Example:

Credit Card

↓

Gateway A — Primary

Gateway B — Secondary

Gateway C — Regional

Board 2 will own that routing.

Payment Transaction State

Although detailed transaction operations come later, Board 1 should use a standardized payment state vocabulary.

Recommended states:

- Initiated
- Pending
- Authorized
- Captured
- Partially Captured
- Failed
- Declined
- Cancelled
- Voided
- Reversed
- Partially Refunded
- Refunded

- Chargeback
- Disputed

This provides consistency across all payment boards.

Multi-Currency Principle

TICVAI should never assume:

Sale Currency = Payment Currency = Settlement Currency.

These can be different.

Example:

Sale

USD

Payment

AED

Settlement

AED

Corporate Reporting

SAR

The data model should therefore retain each currency separately.

Tokenization & PCI Principle

TICVAI should minimize its exposure to raw cardholder data.

Recommended architecture:

Customer

↓

Provider-hosted / certified payment component

↓

Payment Provider / Acquirer

↓

Token

↓

TICVAI

stores the approved reference/token and transaction metadata needed for business operations.

TICVAI should avoid storing sensitive card credentials unless there is a specific, properly certified requirement.

Recurring Payment Configuration

Board 1 should allow a payment method to declare whether it supports recurring transactions.

This becomes important for:

- Membership
- Subscription
- Installment plans

- Recurring services

Example:

Card

Recurring:

Supported.

Cash

Recurring:

Not supported.

The actual recurring transaction orchestration can be handled in the appropriate payment/subscription flow.

Offline / Degraded Capability Flag

Each payment method should expose whether it supports:

- Online only
- Offline/degraded mode
- Store-and-forward where supported
- Not applicable

But the detailed implementation belongs to Boards 2 and 3.

This is especially relevant to your matrix requirement around **payment degraded mode**.

Payment Provider Abstraction

I strongly recommend that does not hard-code TICVAI directly around one provider such as Stripe.

Instead:

TICVAI Payment API

↓

Payment Orchestration Layer

↓

Provider Adapter

↓

Gateway / Acquirer

This means Stripe can be one provider without becoming TICVAI's payment architecture.

This is essential because different clients may use different acquirers in UAE, Saudi Arabia, Europe, or other markets.

Key Integrations

Ticketing

Order/payment requirements.

POS

Card/cash tender availability.

B2C

Online payment methods.

B2B

Credit/invoice payment.

F&B

POS payments.

Retail

POS payments.

Membership

Recurring payment capability.

Wallet

Stored-value payments.

Promotions

Payment-method-related discounts.

Pricing

Transaction amount.

Tax

Tax calculation.

Finance

Accounting and settlement.

Subscription & Licensing

Recurring platform/client billing where applicable.

Board 2

Gateway routing.

Board 3

Terminal/card-present configuration.

AI Capabilities for Board 1

AI should remain primarily **advisory** at this level.

Examples:

"B2C is enabled for AED transactions, but no active payment provider currently supports the configured combination."

"Venue B has no backup digital payment method."

"Cash is enabled for Mobile App although the channel does not support cash collection."

"This payment configuration differs from the tenant standard."

"Disabling Gateway X will leave Kiosk without an available card-processing path."

AI should assist with:

- Configuration validation
- Configuration anomaly detection
- Dependency analysis
- Missing configuration detection
- Impact analysis

- Recommended setup
- Natural-language configuration search

AI should **not autonomously activate financial configuration changes**.

Roles & Permissions

Recommended roles:

- Payment Administrator
- **Finance Manager**
- **Commercial Manager**
- **Revenue Manager**
- POS Administrator
- **E-Commerce Manager**
- **Venue Manager**
- **B2B Manager**
- System Administrator
- Auditor

Sensitive Permissions

Separately control:

- Create payment method
 - Activate/deactivate method
 - Change transaction limit
 - Change channel availability
 - Change currency
 - Configure fee
 - Change provider assignment
 - Approve payment configuration
 - Override configuration
 - Publish configuration
-

Audit Requirements

All sensitive configuration changes must record:

- Tenant
- User
- Role
- Action
- Object
- Old value

- New value
- Date/time
- Approval
- Reason
- Source
- Version

Payment configuration should never be silently modified.

Board 1 Acceptance Criteria

Board 1 shall be considered complete when:

1. TICVAI provides a centralized Payment Command Center.
2. Payment methods can be created and managed centrally.
3. Unlimited/extensible payment-method configuration is supported.
4. Payment methods can be enabled/disabled independently.
5. Payment methods can be assigned by channel.
6. Payment methods can be assigned by venue.
7. Payment methods can be assigned by location/business unit.
8. Ticketing, F&B, Retail and other modules can consume the centralized payment configuration.
9. Payment availability can vary by customer/transaction context.
10. B2B credit/invoice methods can be configured.
11. Gift card/voucher/stored-value methods can be represented.
12. Recurring-payment capability can be identified.
13. Refund capability can be configured per method.
14. Multiple currencies are supported.
15. Selling, payment, settlement and reporting currencies can remain distinct.
16. Payment methods can be restricted by currency.
17. Payment eligibility rules support AND/OR/NOT logic.
18. Minimum and maximum payment amounts can be configured.
19. Payment fees/surcharges can be configured where applicable.
20. Tax treatment of payment fees can be defined.
21. Tenant-level defaults are supported.
22. Venue/location overrides are supported.
23. Channel-level overrides are supported.
24. Configuration inheritance is visible.
25. Effective dates are supported.
26. Configuration versioning is supported.
27. Sensitive changes can require approval.
28. Payment configuration changes are fully audited.
29. Administrators can simulate payment-method availability.
30. Simulation explains why a payment method is unavailable.

31. Publication validation detects missing gateway dependencies.
 32. Publication validation detects unsupported currencies.
 33. Publication validation detects configurations leaving a channel with no valid payment method.
 34. Payment method and gateway are modeled separately.
 35. Standardized transaction states are available across the module.
 36. Offline/degraded capability can be identified without duplicating terminal/gateway implementation.
 37. Provider architecture supports multiple PSPs/acquirers.
 38. TICVAI can minimize handling of raw cardholder data through provider tokenization.
 39. AI can identify configuration risks and inconsistencies without autonomously publishing financial changes.
 40. All configuration respects multi-tenant isolation and TICVAI RBAC/PBAC.
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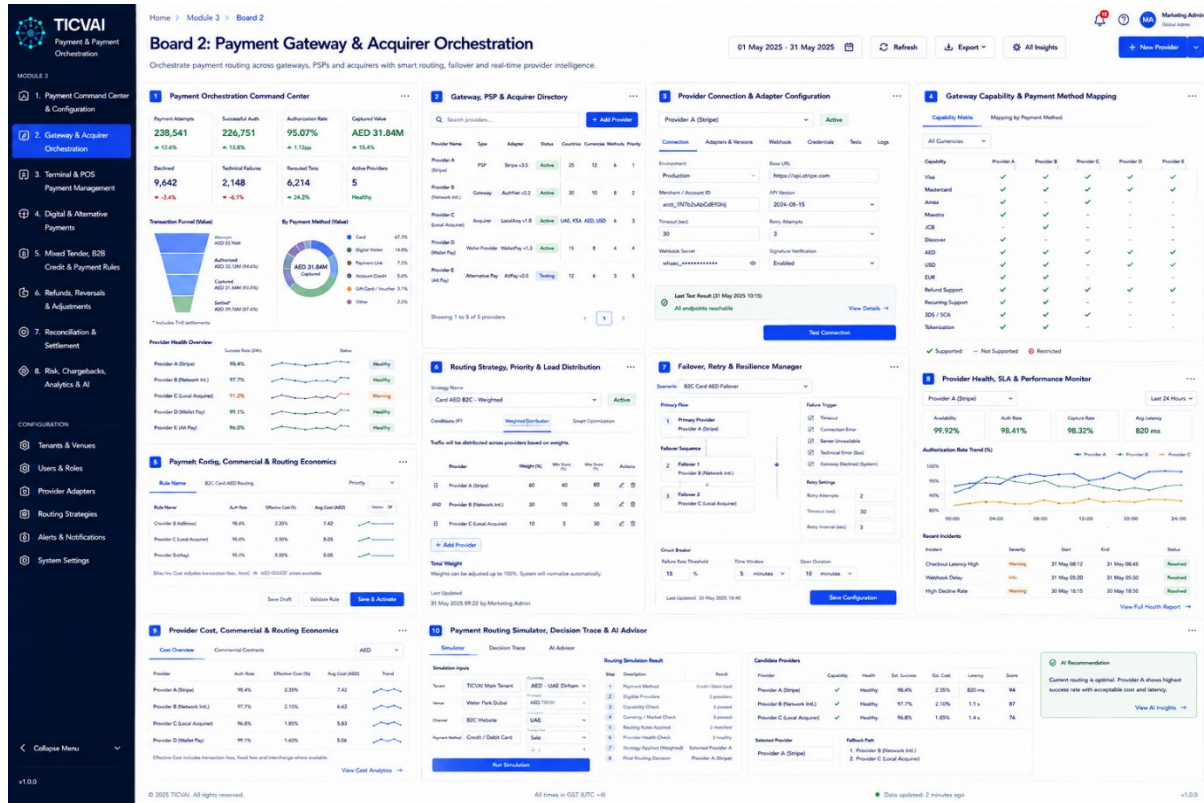
Board 1 — Final 10 Screens

1. **Payment Command Center**
2. **Payment Method Catalogue**
3. **Payment Method Configuration**
4. **Channel & Touchpoint Payment Configuration**
5. **Venue, Location & Business Unit Payment Assignment**
6. **Currency & Payment Currency Configuration**
7. **Payment Eligibility & Availability Rule Builder**
8. **Payment Fees, Surcharges & Commercial Rules**
- 9. Payment Policy, Governance & Approval Manager**
10. **Payment Configuration Simulator & Validation Center**

Board 1 is locked as: *Payment Command Center & Configuration — 10 backend screens.*

Module 3 — Payment & Payment Orchestration

Board 2 — Payment Gateway & Acquirer Orchestration



Board Objective

Board 2 shall provide TICVAI with the centralized **Payment Gateway, PSP & Acquirer Orchestration Layer** responsible for determining **how and through which provider each eligible electronic payment transaction is processed**.

Board 1 determines:

Which payment methods are available to the customer?

Board 2 determines:

Which gateway, PSP or acquirer should process the selected payment, and what happens if that processing path is unavailable or unsuccessful?

This board must allow TICVAI to operate independently of any single payment provider. A client may use Stripe, a local UAE/Saudi acquirer, an international PSP, or multiple providers simultaneously without requiring TICVAI's checkout architecture to be redesigned.

The architecture should support:

TICVAI Payment API → Orchestration Layer → Provider Adapter → Gateway / PSP / Acquirer

Board 2 shall contain 10 backend screens.

Screen 1 — Payment Orchestration Command Center

Purpose

Provide real-time operational visibility across all payment gateways, PSPs and acquirers.

KPI Cards

- Payment Attempts
- Successful Authorizations

- Authorization Rate
- Successful Captures
- Declined Transactions
- Technical Failures
- Routed Transactions
- Rerouted Transactions
- Failover Transactions
- Active Providers
- Provider Incidents
- Average Processing Time

Transaction Value

Display:

- Attempted Value
- Authorized Value
- Captured Value
- Failed Value
- Rerouted Value

Provider Performance

Provider	Transactions	Value	Success	Latency	Status
Provider A	62K	AED 8.4M	98.4%	820ms	Healthy
Provider B	31K	AED 4.2M	97.7%	1.1s	Healthy
Provider C	12K	AED 1.7M	91.2%	2.4s	Warning

Operational Alerts

Examples:

- Authorization rate degradation
- Provider timeout increase
- Provider unavailable
- Routing rule conflict
- Currency unsupported
- Credential/certificate nearing expiry
- Failover activated

Screen 2 — Gateway, PSP & Acquirer Directory

Purpose

Maintain the centralized directory of payment-processing providers connected to TICVAI.

Provider Types

Support:

- Payment Gateway
- PSP
- Acquirer
- Processor
- Bank
- Wallet Provider
- Alternative Payment Provider

Provider Record

Each provider should contain:

- Provider ID
- Provider Name
- Provider Type
- Adapter
- Countries/Markets
- Supported Currencies
- Supported Payment Methods
- Card-Present / Card-Not-Present
- Recurring Support
- Refund Support
- Tokenization Support
- 3DS Capability
- **Status**

Status

- Draft
- Testing
- Active
- Degraded
- Suspended
- Maintenance
- Retired

Important

The directory represents **processing capabilities**, not customer-facing payment methods.

Screen 3 — Provider Connection & Adapter Configuration

Purpose

Configure how TICVAI technically connects to each payment provider.

Connection Configuration

Support:

- Environment

- Sandbox
- Production
- API endpoint reference
- Merchant/account reference
- Authentication method
- Webhook configuration
- Callback configuration
- Timeout settings
- Retry configuration

Adapter Version

Display:

- TICVAI Adapter
- Adapter Version
- Provider API Version
- Deployment Version
- Last Certification/Test
- **Status**

Credential Management

The UI should reference securely stored credentials rather than display sensitive credentials.

Show:

- Credential status
- Last rotation
- Expiry
- Certificate expiry
- Owner

Never expose full:

- API secrets
- Private keys
- Card credentials

Test Connection

Authorized administrator can execute:

Test Provider Connection

Result:

- Authentication ✓
- API Reachability ✓
- Authorization Endpoint ✓
- Refund Endpoint ✓
- Webhook ✓

Screen 4 — Gateway Capability & Payment Method Mapping

Purpose

Map Board 1 payment methods to providers capable of processing them.

Example

Credit / Debit Card

Mapped to:

- Provider A
- Provider B
- Provider C

Digital Wallet

Mapped to:

- Provider A
- Provider D

Payment Link

Mapped to:

- Provider A

Capability Matrix

Capability	A	B	C
Visa	✓	✓	✓
Mastercard	✓	✓	✓
Amex	✓	—	✓
AED	✓	✓	✓
USD	✓	✓	—
Recurring	✓	✓	—
Refund	✓	✓	✓
Tokenization	✓	✓	—
3DS	✓	✓	✓

Mapping Dimensions

Allow provider capability to vary by:

- Tenant
- Legal entity
- Merchant account
- Venue
- Country
- Currency
- Channel
- Payment method
- Transaction type

This prevents the routing engine from selecting an incompatible provider.

Screen 5 — Payment Routing Rule Builder

Purpose

Configure the rules that determine which provider should receive each payment transaction.

Rule Inputs

Support:

- Tenant
- Venue
- Legal entity
- Channel
- Payment method
- Card/payment type
- Currency
- Country/market
- Transaction amount
- Transaction type
- Merchant account
- Customer type where appropriate
- Provider health

Example Rule

IF

Channel = B2C

AND

Currency = AED

AND

Payment Method = Card

THEN

Route:

Provider A

Priority:

1

Fallback:

Provider B

Rule Logic

Support:

- AND
- OR
- NOT
- Include
- Exclude

Routing Methods

Support:

- Fixed provider
- Priority-based
- Weighted distribution
- Cost-based
- Performance-based
- Availability-based
- Smart/dynamic routing

Screen 6 — Routing Strategy, Priority & Load Distribution

Purpose

Manage how payment traffic is distributed when several valid providers are available.

Strategies

Primary / Backup

Provider A — Primary

Provider B — Backup

Weighted

Provider A — 70%

Provider B — 30%

Lowest Processing Cost

Route to the lowest-cost eligible provider.

Highest Authorization Rate

Prefer the provider with the strongest recent approval performance.

Lowest Latency

Prefer the fastest eligible provider.

Balanced Optimization

Combine:

- Authorization probability
- Cost
- Latency
- Provider health
- Commercial commitments

Traffic Allocation

Example:

AED Card / B2C

Provider A: 60%

Provider B: 30%

Provider C: 10%

Commercial Commitment Controls

Where contractual commitments exist:

- Minimum transaction share

- Maximum share
- Volume threshold
- Effective period

Screen 7 — Failover, Retry & Resilience Manager

Purpose

Maintain payment availability when a provider experiences a technical problem.

This is one of the most important screens in the Payment Orchestration module.

Failover Rules

Configure:

- Primary provider
- Secondary provider
- Tertiary provider
- Trigger condition
- Retry limit
- Timeout
- Cooldown
- Recovery behavior

Example

Provider A

↓

Timeout

↓

Retry permitted?

↓

No / Retry exhausted

↓

Provider B

Failure Classification

Differentiate:

Technical Failure

Examples:

- Timeout
- Network error
- Provider unavailable

May allow rerouting.

Customer / Issuer Decline

Examples:

- Insufficient funds
- Invalid card

- Issuer decline

Should **not automatically be retried through multiple providers** merely to bypass a legitimate decline.

This distinction is critical.

Circuit Breaker

If:

Provider A technical failure rate > configured threshold

Then:

Temporarily remove Provider A from routing.

Recovery

Provider can return through:

- Automatic health recovery
- Manual approval
- Controlled percentage rollout

Screen 8 — Provider Health, SLA & Performance Monitor

Purpose

Monitor each provider's technical and transactional health.

Health Metrics

- Availability
- Authorization rate
- Capture rate
- Decline rate
- Technical error rate
- Timeout rate
- Average latency
- P95/P99 latency where available
- Webhook delay
- Refund API health

Health State

- Healthy
- Warning
- Degraded
- Critical
- Offline
- Maintenance

SLA Monitoring

Configure:

- Expected uptime
- Maximum latency
- Minimum authorization benchmark

- Technical failure threshold

Example Alert

Provider C

Authorization Rate:

92.1%

Normal Baseline:

97.6%

Change:

–5.5 pp

Status:

Investigate

Trend Analysis

Compare:

- Today
- Yesterday
- 7 Days
- 30 Days
- Historical baseline

Screen 9 — Provider Cost, Commercial & Routing Economics

Purpose

Allow TICVAI to understand the commercial cost of routing transactions through different providers.

Cost Configuration

Where available, support:

- Fixed transaction fee
- Percentage fee
- Authorization fee
- Capture fee
- Refund fee
- Cross-border fee
- Currency conversion cost
- Minimum fee
- Other contracted processing costs

Example

Provider	Auth Rate	Avg Cost	Latency
A	98.4%	2.35%	0.8s
B	97.9%	2.10%	1.0s
C	96.8%	1.85%	1.4s

The cheapest provider is therefore **not necessarily the best route**.

Effective Processing Cost

Calculate:

- Provider fees
- Successful transaction cost
- Failure cost where modeled
- Routing cost
- Cost by channel
- Cost by currency
- Cost by market

Optimization Objective

Administrator may define:

- Maximize Authorization
- Minimize Cost
- Minimize Latency
- Balanced
- Custom Governed Strategy

Screen 10 — Payment Routing Simulator, Decision Trace & AI Advisor

Purpose

Allow administrators to test exactly how a payment will be routed before deploying routing changes.

Simulation Inputs

Select:

- Tenant
- Venue
- Legal Entity
- Channel
- Payment Method
- Currency
- Amount
- Market
- Transaction Type
- Provider health scenario

Example

Transaction:

AED 720

Channel:

B2C

Method:

Credit Card

Venue:

Water Park Dubai**Candidate Providers****Provider A**

Capability ✓
Currency ✓
Channel ✓
Health ✓
Routing Rule ✓

Score: **94**

Provider B

Capability ✓
Currency ✓
Channel ✓
Health ✓

Score: **87**

Provider C

Capability ✓
Currency ✓
Health: Degraded

Score: **61**

Final Route**Provider A**

Fallback:

Provider B**Routing Decision Trace**

The screen should show:

Payment Request

↓

Payment Method

↓

Eligible Providers

↓

Capability Check

↓

Currency / Market Check

↓

Routing Rules

↓

Provider Health

↓

Commercial Constraints

↓

Routing Strategy

↓

Selected Provider

↓

Fallback Path

This gives TICVAI full explainability.

AI Routing Advisor

AI should analyze historical and real-time provider performance and make recommendations.

Examples:

Provider B currently has a 1.8 percentage-point higher authorization rate for AED B2C transactions than Provider A.

Moving 20% of traffic to Provider B is estimated to recover 640 successful transactions per month.

Or:

Provider C is 14% cheaper but currently has materially higher timeout rates. No routing change is recommended.

AI Can Recommend

- Routing weight changes
- Provider priority changes
- Temporary traffic reduction
- Failover activation
- Provider investigation
- Commercial optimization
- Routing experiment

AI Should Not

Autonomously:

- Add a provider
 - Change credentials
 - Bypass approvals
 - Route to an uncertified provider
 - Override provider capability
 - Ignore compliance rules
-

Smart Routing Architecture

I recommend to implement routing as a deterministic eligibility layer followed by an optimization layer.

Step 1 — Hard Eligibility

Provider must support:

- Payment method
- Currency
- Market
- Channel
- Transaction type
- Merchant account

↓

Step 2 — Provider Availability

Provider must be operational.

↓

Step 3 — Commercial / Contractual Rules

Apply:

- Routing commitments
- Provider restrictions
- Tenant rules

↓

Step 4 — Optimization

Choose among remaining providers based on:

- Authorization performance
- Cost
- Latency
- Reliability
- Configured objective

↓

Selected Provider

This ensures AI cannot recommend an invalid processing route.

Payment Provider Adapter Architecture

Development team should create a standard TICVAI payment-provider interface.

Conceptually:

TICVAI Payment Orchestration

↓

Standard Payment Adapter Contract

Functions such as:

- Authorize
- Capture
- Sale
- Void
- Refund
- Query Status

- Tokenize where applicable
- Create Payment Link where supported
- Handle Callback/Webhook

↓

Provider Adapters

- Provider A Adapter
- Provider B Adapter
- Provider C Adapter
- Stripe Adapter
- Future Provider Adapter

This avoids putting provider-specific logic throughout TICVAI.

Idempotency

This is a critical technical requirement.

Every payment operation should support an appropriate **idempotency mechanism** so network retries do not accidentally create duplicate payments.

Example:

Checkout sends:

Payment Request 847291

Network timeout occurs.

Checkout retries.

TICVAI must determine:

Is this a new transaction or the same payment request?

and avoid unintended duplicate capture.

Transaction Status Synchronization

TICVAI must not rely only on the immediate browser/POS response.

Transaction status may also be updated by:

- Provider response
- Webhook
- Callback
- Status inquiry
- Reconciliation

Example:

Customer browser times out.

Provider actually captured:

AED 720

TICVAI should later reconcile the payment state correctly rather than automatically charging the customer again.

Webhook & Callback Governance

Each provider integration should support:

- Event verification
- Signature validation
- Duplicate event protection
- Event timestamp
- Event processing status
- Retry handling
- Failed webhook queue
- **Audit**

This will be important for online and asynchronous payment methods.

Routing vs Failover

These should be treated separately.

Routing

Selects the best provider **before** attempting payment.

Failover

Determines what happens **after a technical processing failure**.

Example:

Routing:

Provider A selected.

Provider A times out.

Failover:

Provider B may be attempted if permitted.

This distinction should appear clearly in the backend UI.

Degraded Mode

The matrix specifically requires degraded-mode capability.

Board 2 should define gateway-side resilience behavior such as:

- Provider unavailable
- Network degraded
- API timeout
- Provider maintenance

The detailed **physical terminal offline/store-and-forward behavior** belongs to Board 3.

So:

Board 2

Gateway/API resilience.

Board 3

Terminal/card-present resilience.

Multi-Tenant & Multi-Merchant Architecture

TICVAI should support:

Tenant



Legal Entity



Merchant Account



Venue / Channel



Provider

A client may therefore use:

Provider A for UAE.

Provider B for Saudi Arabia.

Provider C for Europe.

while remaining one TICVAI tenant.

Merchant Account Mapping

A single provider may have several merchant accounts.

Example:

Provider A

- Water Park Merchant Account
- Aquarium Merchant Account
- Retail Merchant Account
- F&B Merchant Account

Routing must therefore select not only:

Provider A

but potentially:

Provider A + Merchant Account X

This is important for settlement and reconciliation later.

Provider Token Portability Principle

Where technically possible, TICVAI should avoid unnecessary lock-in to one provider's token model.

However, payment tokens are often provider-specific.

Therefore TICVAI should maintain a normalized internal payment-instrument reference while retaining provider-specific token references securely.

This will support future multi-provider architecture without pretending all provider tokens are interchangeable.

Relationship with Board 1

Board 1

Customer can pay by:

Credit Card

Board 2

Credit Card can route through:

- Provider A
- Provider B
- Provider C

Result

Board 1 controls **availability**.

Board 2 controls **processing route**.

Relationship with Board 3

Board 3 will handle card-present processing:

POS

↓

Terminal

↓

Terminal/Payment Server

↓

Acquirer

Board 2 can still provide the provider/acquirer configuration and orchestration principles, while Board 3 handles:

- Terminal inventory
- Device assignment
- EMV
- Terminal status
- Card-present workflow
- Terminal degraded mode

This avoids duplication.

Relationship with Board 7 — Reconciliation

Board 2 knows:

Which provider processed the transaction.

Board 7 will determine:

Did the provider settle the correct amount?

Therefore every routed transaction must retain:

- Provider
- Merchant account
- Provider transaction ID
- Routing rule

- Routing decision
- Attempt number
- Failover history

for later reconciliation.

AI Architecture

AI should operate **above hard routing constraints**.

Correct architecture:

Hard Eligibility

↓

Commercial Rules

↓

Provider Health

↓

AI / Optimization Score

↓

Final Route

Not:

AI chooses any gateway it wants.

This makes the solution safer and auditable.

Roles & Permissions

Recommended roles:

- Payment Administrator
- **Payment Operations Manager**
- **Finance Manager**
- **E-Commerce Manager**
- POS Administrator
- Integration Administrator
- Technical Administrator
- **Risk Manager**
- System Administrator
- Auditor

Sensitive Permissions

Separately control:

- Add provider
- Activate provider
- Change provider connection

- Change merchant account
 - Change routing priority
 - Change routing weight
 - Configure failover
 - Disable provider
 - Change cost information
 - Activate smart routing
 - Publish routing strategy
-

Audit Requirements

Record all routing configuration changes including:

- User
- Role
- Tenant
- Provider
- Merchant account
- Previous configuration
- New configuration
- Date/time
- Approval
- Effective date
- Reason
- Version

For individual payment decisions, retain sufficient decision metadata to reconstruct why a route was selected.

Board 2 Acceptance Criteria

Board 2 shall be considered complete when:

1. TICVAI provides a centralized Payment Orchestration Command Center.
2. Multiple gateways, PSPs and acquirers are supported.
3. Providers are abstracted through standardized adapters.
4. Sandbox and production environments are supported.
5. Provider credentials are handled securely and are not exposed through the UI.
6. Provider connection health can be tested.
7. Payment methods can map to multiple providers.
8. Provider capabilities can be configured.
9. Currency capability can be configured.
10. Market/country capability can be configured.
11. Channel capability can be configured.
12. Card-present/card-not-present capability can be distinguished.

13. Merchant accounts can be mapped independently of providers.
14. Routing rules can be created.
15. Routing supports priority-based selection.
16. Routing supports weighted distribution.
17. Routing can consider provider health.
18. Routing can consider authorization performance.
19. Routing can consider processing cost.
20. Routing can consider latency.
21. Contractual traffic constraints can be represented.
22. Failover providers can be configured.
23. Technical failures are distinguishable from issuer/customer declines.
24. Legitimate declines are not blindly retried through multiple providers.
25. Circuit-breaker behavior is supported.
26. Provider recovery can be controlled.
27. Provider SLA and health are monitored.
28. Authorization-rate degradation can generate alerts.
29. Provider latency can be monitored.
30. Provider cost information can support routing optimization.
31. Routing decisions can be simulated before deployment.
32. Simulation displays candidate providers and rejection reasons.
33. Every route can provide an explainable decision trace.
34. AI can recommend routing optimization without bypassing hard constraints.
35. Idempotency is supported for payment operations.
36. Asynchronous callbacks/webhooks are governed and auditable.
37. Payment status can recover from uncertain/time-out states.
38. Gateway-side degraded conditions are supported without duplicating terminal offline functionality.
39. Routing data is retained for downstream reconciliation and settlement.
40. All provider and routing configuration respects TICVAI multi-tenancy, RBAC/PBAC, approval and audit requirements.

Board 2 — Final 10 Screens

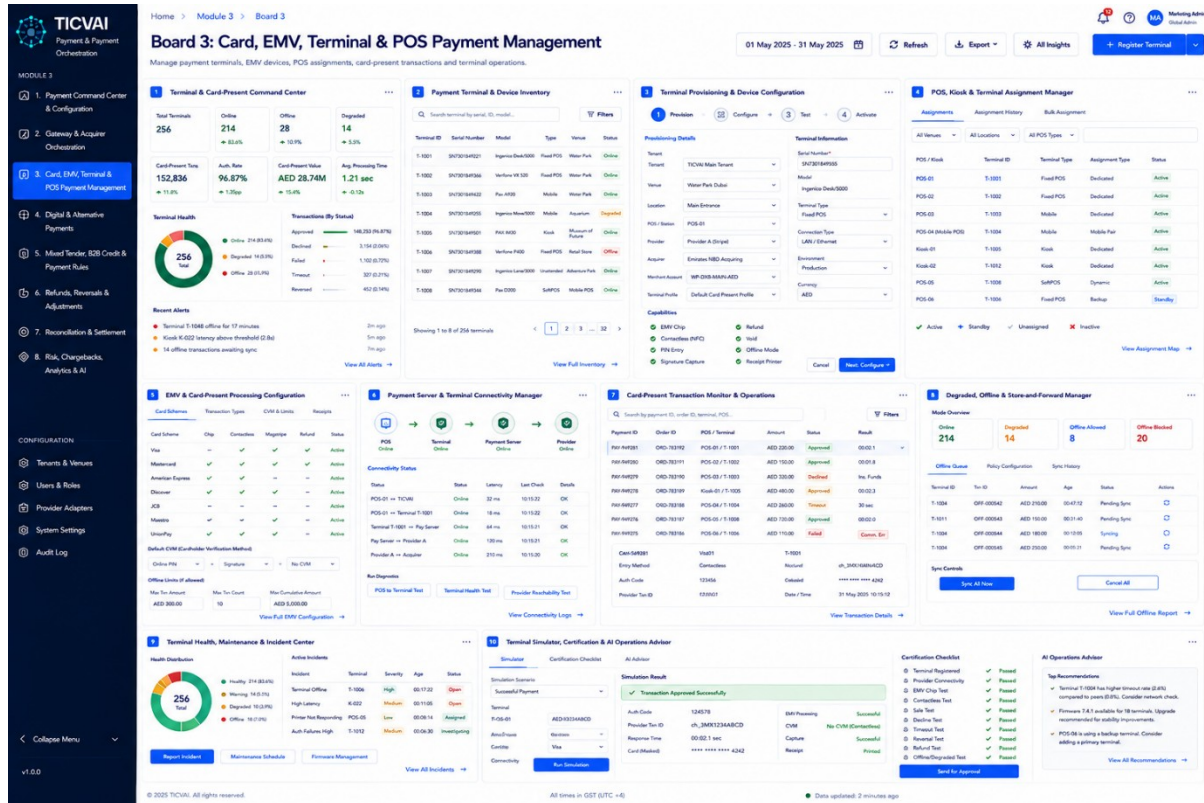
1. **Payment Orchestration Command Center**
2. **Gateway, PSP & Acquirer Directory**
3. **Provider Connection & Adapter Configuration**
4. **Gateway Capability & Payment Method Mapping**
5. **Payment Routing Rule Builder**
6. **Routing Strategy, Priority & Load Distribution**
7. **Failover, Retry & Resilience Manager**
8. **Provider Health, SLA & Performance Monitor**
9. **Provider Cost, Commercial & Routing Economics**

10. Payment Routing Simulator, Decision Trace & AI Advisor

Board 2 is locked as: *Payment Gateway & Acquirer Orchestration — 10 backend screens.*

Module 3 — Payment & Payment Orchestration

Board 3 — Card, EMV, Terminal & POS Payment Management



Board Objective

Board 3 shall provide TICVAI with the centralized backend for managing **card-present payments, EMV devices, payment terminals, terminal-to-POS assignments, payment-server connectivity, transaction execution, terminal health, and degraded/offline operations.**

The separation from the previous boards is:

Board 1 — Payment Configuration

Determines **which payment methods are allowed.**

Board 2 — Gateway & Acquirer Orchestration

Determines **which provider/acquirer processing path should be used.**

Board 3 — Terminal & POS Payment Management

Determines:

Which physical payment device processes the card-present transaction, how it communicates with TICVAI/POS, and how terminal operations are controlled and monitored.

This board must support fixed POS, mobile/floating POS, kiosks and other approved attended/unattended card-present environments.

Board 3 shall contain 10 backend screens.

Screen 1 — Terminal & Card-Present Command Center

Purpose

Provide real-time operational visibility over all card-present payment infrastructure.

KPI Cards

- Total Terminals

- Online Terminals
- Offline Terminals
- Degraded Terminals
- Active POS Assignments
- Card-Present Transactions
- Authorization Rate
- Card-Present Value
- Failed Transactions
- Average Processing Time
- Terminal Alerts
- Offline/Pending Transactions

Terminal Health

Break down:

- Online
- Warning
- Degraded
- Offline
- Maintenance
- Disabled

Transaction Performance

Analyze by:

- Venue
- Location
- Terminal
- POS
- Acquirer
- Payment method
- Card scheme
- Device model

Alerts

Examples:

Terminal T-1048 offline for 17 minutes.

Kiosk terminal K-022 has authorization latency above threshold.

14 degraded-mode transactions awaiting synchronization.

Screen 2 — Payment Terminal & Device Inventory

Purpose

Maintain the centralized inventory of all payment terminals connected to TICVAI.

Terminal Record

For every device retain:

- Terminal ID
- Serial Number
- Device Model
- Manufacturer
- Terminal Type
- Provider
- Acquirer
- Merchant Account
- Venue
- Location
- POS Assignment
- Connection Type
- Software/Firmware Version
- **Status**

Terminal Types

Support:

- Fixed POS terminal
- Mobile terminal
- Integrated terminal
- Standalone terminal
- Kiosk terminal
- Unattended terminal
- SoftPOS where supported
- Other certified devices

Lifecycle

Received

→ **Provisioning**

→ **Testing**

→ **Active**

→ **Maintenance**

→ **Suspended**

→ **Retired**

Asset Controls

Support:

- Device owner
- Installation date
- Warranty
- Last service
- Replacement date
- Notes

Screen 3 — Terminal Provisioning & Device Configuration

Purpose

Configure and activate a physical terminal for TICVAI use.

Provisioning Configuration

Select:

- Tenant
- Legal Entity
- Venue
- Location
- POS
- Provider
- Acquirer
- Merchant Account
- Terminal Profile

Terminal Configuration

Support:

- Terminal ID
- Device reference
- Environment
- Currency
- Country
- Language
- Receipt configuration
- Timeout
- Connection profile

Capabilities

Display whether the terminal supports:

- EMV Chip
- Contactless
- NFC
- Magnetic stripe where permitted
- PIN
- Signature where applicable
- Refund
- Void
- Preauthorization
- Offline/degraded capability

Provisioning Validation

Before activation verify:

- ✓ Device registered
 - ✓ Merchant mapping exists
 - ✓ Provider connected
 - ✓ Currency supported
 - ✓ POS assigned
 - ✓ Required capabilities available
-

Screen 4 — POS, Kiosk & Terminal Assignment Manager

Purpose

Control which terminal is connected or assigned to each TICVAI selling device.

Assignment Hierarchy

Tenant

↓

Venue

↓

Location

↓

POS / Kiosk

↓

Payment Terminal

Example

Water Park — Main Entrance

POS-01 → Terminal T-1001

POS-02 → Terminal T-1002

POS-03 → Terminal T-1003

Assignment Types

Support:

Dedicated

One terminal permanently assigned to one POS.

Shared

Approved terminals shared between multiple selling stations.

Dynamic

Terminal selected dynamically according to approved integration capability.

Mobile

Mobile terminal assigned to floating/mobile POS.

Assignment Controls

- Primary terminal
- Backup terminal
- Effective date
- Shift restrictions

- Location restriction

- **Status**

Conflict Detection

Prevent:

- Invalid duplicate assignment
- Wrong merchant account
- Unsupported venue
- Wrong currency
- Incompatible terminal/POS

Screen 5 — EMV & Card-Present Processing Configuration

Purpose

Configure card-present transaction capabilities without exposing sensitive cardholder data.

Supported Entry Methods

Where provider/device certified:

- EMV Chip
- Contactless EMV
- NFC Wallet
- PIN
- Magnetic stripe where allowed
- Manual entry only where specifically permitted

Card Schemes

Examples:

- Visa
- Mastercard
- American Express
- Other provider-supported schemes

Transaction Types

Configure:

- Sale
- Authorization
- Capture
- Void
- Refund
- Reversal
- Preauthorization where supported
- Completion where supported

Cardholder Verification

Based on provider/terminal capability:

- Online PIN

- Offline PIN
- Signature
- No CVM
- Device verification

Important Architecture Rule

TICVAI should store business/payment references such as:

- Payment transaction ID
- Provider transaction ID
- Terminal ID
- Authorization result
- Masked PAN if supplied/allowed
- Card brand
- Token/reference

but should avoid storing prohibited sensitive authentication/card data.

Screen 6 — Payment Server & Terminal Connectivity Manager

Purpose

Monitor and configure the communication path between TICVAI POS, payment terminal, payment server and payment provider.

Architecture Example

TICVAI POS

↓

Terminal Integration Layer

↓

Payment Terminal

↓

Payment Server / Provider

↓

Acquirer

Connection Modes

Depending on integration:

- Local/LAN
- Cloud
- API
- Payment server
- Terminal middleware
- Mobile
- Other certified integration

Connectivity Monitor

Show:

- POS ↔ Terminal
- Terminal ↔ Payment Server
- Payment Server ↔ Provider
- Provider ↔ Acquirer

Health Indicators

- Connected
- Slow
- Intermittent
- Offline
- Unknown

Diagnostics

Authorized users can run:

- Connectivity test
- Terminal ping/health test where supported
- Provider reachability
- Merchant validation
- Configuration validation

Alert

POS-17 can communicate with TICVAI but its payment terminal cannot reach the payment service.

This distinction is operationally very important.

Screen 7 — Card-Present Transaction Monitor & Operations

Purpose

Provide operations teams with real-time visibility into card-present payment attempts.

Transaction Search

Search/filter by:

- TICVAI Payment ID
- Order
- POS
- Terminal
- Venue
- Provider
- Acquirer
- Date/time
- Amount
- Currency
- **Status**
- Operator

Transaction Record

Display:

- Payment ID
- Order ID
- Terminal
- POS
- Amount
- Currency
- Provider
- Merchant account
- Card scheme
- Entry method
- Authorization result
- Processing duration
- **Status**

Status Examples

- Initiated
- Awaiting Terminal
- Card Presented
- Processing
- Authorized
- Captured
- Declined
- Failed
- Timed Out
- Reversal Pending
- Reversed

Important

Do not show prohibited sensitive card data.

Screen 8 — Degraded, Offline & Store-and-Forward Manager

Purpose

Manage card-present operations when normal connectivity is unavailable.

This directly addresses the **degraded-mode requirement** in the payment matrix.

Important Principle

Offline payment behavior must depend on:

- Terminal capability
- Provider capability
- Acquirer rules

- Merchant configuration
- Risk policy

TICVAI must not assume every terminal can accept offline payments.

Modes

Support representation of:

- Normal Online
- Degraded
- Offline Allowed
- Store-and-Forward
- Offline Blocked

Offline Policy

Configure where supported:

- Maximum transaction value
- Maximum offline transactions
- Maximum cumulative value
- Allowed terminal
- Allowed venue
- Allowed product/channel
- Time window
- Synchronization requirement

Example

Venue

Water Park

Offline Maximum

AED 300 / transaction

Maximum Cumulative Exposure

AED 5,000

Synchronization

Immediately when connection returns.

Offline Queue

Show:

- Terminal
- Transaction
- Amount
- Timestamp
- Offline reason
- Retry status
- Synchronization state

Screen 9 — Terminal Health, Maintenance & Incident Center

Purpose

Provide centralized lifecycle monitoring and operational maintenance of terminal hardware.

Device Health

Monitor where technically available:

- Connectivity
- Last heartbeat
- Software version
- Firmware
- Battery for mobile devices
- Network
- Last transaction
- Error count

Incident Types

- Terminal Offline
- Connection Failure
- Payment Server Failure
- Device Error
- Printer Issue
- Firmware Issue
- Configuration Mismatch
- Excessive Declines
- Excessive Timeouts

Incident Workflow

Detected

→ Assigned

→ Investigating

→ Resolved

→ Closed

Maintenance

Support:

- Planned maintenance
- Firmware upgrade
- Device replacement
- Terminal reconfiguration
- Certification update

Fleet View

Example:

Venue	Total Health		Warning	Offline
Water Park	42	39	2	1

Venue	Total	Healthy	Warning	Offline
Aquarium	18	18	0	0
Retail	31	28	2	1

Screen 10 — Terminal Simulator, Certification & AI Operations Advisor

Purpose

Provide a controlled environment to validate terminal configuration and payment flows before production activation.

Simulation Inputs

Select:

- Tenant
- Venue
- POS
- Terminal
- Provider
- Merchant account
- Currency
- Transaction amount
- Transaction type
- Connectivity state

Test Scenarios

Support simulation/test flows for:

- Successful payment
- Card decline
- Terminal timeout
- Provider timeout
- Lost connection
- Reversal
- Duplicate request
- Offline/degraded operation
- Recovery/synchronization

Example

Scenario

POS: POS-04

Terminal: T-1028

Amount: AED 720

Connection:

Normal

Result

POS Configuration ✓
Terminal Assignment ✓
Merchant Mapping ✓
Currency ✓
Provider Connectivity ✓
EMV Capability ✓

Ready for Production

Certification Checklist

Before activation:

- Terminal registered
- Correct firmware
- Correct merchant account
- Payment provider mapping
- EMV capability validated
- Contactless validated
- Sale tested
- Decline tested
- Timeout tested
- Reversal tested
- Refund capability verified
- Offline/degraded behavior validated where applicable

Deployment Status

Certification Passed

→ **Send for Approval**

→ **Activate Terminal**

AI Operations Advisor

AI can identify operational patterns such as:

Terminal T-1044 has experienced 3.7× more communication failures than terminals of the same model.

POS terminals using firmware 7.4.1 have a significantly higher timeout rate.

Kiosk K-08 should be inspected before peak operating hours.

Water Park Gate POS cluster has no healthy backup terminal.

AI Can Recommend

- Device inspection
- Terminal reassignment
- Firmware investigation
- Connectivity investigation
- Backup-terminal deployment
- Preventive maintenance

- Configuration correction

AI should not autonomously change sensitive terminal/payment configuration.

Card-Present Transaction Flow

The standard flow should conceptually be:

POS Creates Payment Request

↓

TICVAI Payment Service

Creates payment transaction + idempotency reference.

↓

Terminal Assignment

Determine correct device.

↓

Terminal

Customer presents card/device.

↓

EMV Processing

↓

Payment Provider / Acquirer

↓

Authorization Response

↓

TICVAI Payment State Updated

↓

POS Receives Result

↓

Order Completion

This transaction flow must be independent enough that a POS UI does not directly contain provider-specific logic.

Timeout / Uncertain Transaction Handling

This is critical.

Example:

Customer taps card.

↓

Terminal sends transaction.

↓

POS loses connection before receiving response.

The POS must **not simply charge again**.

TICVAI should first determine:

What happened to the original payment?

Possible state:

Payment Status Unknown / Pending Verification

Then query:

- Terminal
- Payment server
- Provider

before deciding whether a new attempt is safe.

Reversal Handling

Example:

Authorization succeeds at provider.

But:

Order completion fails.

The system may need to initiate:

Reversal / Void

depending on provider capability and transaction state.

Board 3 should expose the card-present operational state.

Board 6 will own the broader refund/reversal business workflow.

Duplicate Payment Prevention

Each payment request should have a unique reference.

Example:

PAY-847291

Retrying the same request should not automatically create:

PAY-847292

and charge again.

Use:

- Payment ID
- Order ID
- Idempotency reference
- Provider transaction ID
- Terminal transaction reference

to prevent duplicate card-present charges.

POS Operator Experience

The POS should receive clear states.

Blue / Processing

Present card.

Green / Approved

Payment successful.

Red / Declined

Payment declined.

Yellow / Attention

Payment status being verified. Do not retry.

The yellow state is particularly important for avoiding duplicate transactions.

Terminal Receipt Configuration

Where applicable, support configuration of:

- Merchant receipt
- Customer receipt
- Printed
- Digital
- Combined TICVAI receipt
- Terminal-only receipt

The transaction should retain relevant receipt/reference identifiers.

Multi-Merchant Terminal Architecture

A TICVAI tenant may have different legal entities or merchant accounts.

Example:

Water Park POS

→ Water Park Merchant Account.

Retail POS

→ Retail Merchant Account.

F&B POS

→ F&B Merchant Account.

Terminal assignment must validate the correct merchant relationship.

Otherwise, transactions may be processed successfully but settle to the wrong entity.

Mobile / Floating POS

Board 3 must support TICVAI's mobile/floating POS architecture.

Example:

Operator logs into:

Mobile POS 14

↓

Assigned to:

Water Park

↓

Pairs with:

Mobile Terminal 28

↓

Shift starts.

At shift end:

Terminal becomes available for reassignment.

Controls

Support:

- Pair
 - Unpair
 - Temporary assignment
 - Operator
 - Shift
 - Venue
 - Location
 - Assignment history
-

Kiosk Payments

For unattended kiosks, terminal configuration should additionally identify:

- Unattended status
- Supported card interfaces
- Contactless
- Receipt capability
- Device recovery state
- Remote health

Kiosk UI remains part of the POS/Kiosk module.

Board 3 owns the payment-device layer.

PCI & Security Boundary

Development team should design Board 3 under a principle of **minimum card-data exposure**.

TICVAI should not unnecessarily receive or store:

- Full PAN
- CVV/CVC
- PIN
- Sensitive EMV authentication data

Prefer:

Terminal / Certified Payment Component

↓

Payment Provider

↓

TICVAI receives:

- Token/reference
 - Masked payment information where permitted
-

- Authorization result
- Transaction ID
- Card brand
- Terminal ID
- Business metadata

The exact PCI scope will depend on the final integration architecture and providers selected.

Relationship with Board 1

Board 1:

Card is allowed at POS.

Board 3:

POS-01 uses Terminal T-1001.

Relationship with Board 2

Board 2:

Provider A / Acquirer A processes this transaction.

Board 3:

Terminal T-1001 communicates through the approved Provider A integration.

Relationship with Board 6

Board 3 identifies:

Payment authorization/capture/reversal state.

Board 6 will manage:

- Refund request
- Partial refund
- Reversal workflows
- Void
- Payment adjustments

Therefore Board 3 should not duplicate the full refund-management system.

Relationship with Board 7

Board 3 records:

- Terminal ID
- POS
- Merchant account
- Provider transaction
- Authorization
- Amount

Board 7 uses this information for:

Reconciliation & Settlement.

Terminal Event Architecture

Recommended standardized events include:

- TERMINAL_ONLINE
- TERMINAL_OFFLINE
- TERMINAL_ASSIGNED
- TERMINAL_UNASSIGNED
- PAYMENT_STARTED
- CARD_PRESENTED
- PAYMENT_AUTHORIZED
- PAYMENT_DECLINED
- PAYMENT_FAILED
- PAYMENT_TIMEOUT
- PAYMENT_STATUS_UNKNOWN
- REVERSAL_STARTED
- REVERSAL_COMPLETED
- OFFLINE_PAYMENT_STORED
- OFFLINE_PAYMENT_SYNCED

These events can feed:

- Operations
- Monitoring
- **Audit**
- Analytics
- AI

Roles & Permissions

Recommended roles:

- Payment Administrator
- POS Administrator
- Terminal Administrator
- **Venue Manager**
- Payment Operations
- IT Support
- Integration Administrator
- **Finance Manager**
- System Administrator
- Auditor

Sensitive Permissions

Separately control:

- Register terminal
 - Provision terminal
 - Assign terminal
 - Change merchant account
 - Enable offline mode
 - Change offline limit
 - Disable terminal
 - Change provider mapping
 - Approve certification
 - Activate production terminal
-

Audit Requirements

Record:

- Terminal
- Serial/device reference
- Tenant
- Venue
- POS
- Merchant account
- Provider
- User
- Action
- Old value
- New value
- Timestamp
- Approval
- Reason

Also retain complete terminal assignment history.

Board 3 Acceptance Criteria

Board 3 shall be considered complete when:

1. TICVAI provides a centralized Terminal & Card-Present Command Center.
2. Payment terminals can be inventoried.
3. Terminal lifecycle status is managed.
4. Fixed terminals are supported.
5. Mobile/floating terminals are supported.
6. Kiosk/unattended terminals can be represented.
7. Terminal capabilities are configurable.
8. EMV chip capability can be identified.

9. Contactless/NFC capability can be identified.
10. Card-present transaction types can be configured.
11. Terminals can be provisioned by tenant/venue/location.
12. Merchant accounts can be mapped to terminals.
13. Terminals can be assigned to POS devices.
14. Primary and backup terminal assignments are supported.
15. Mobile terminal pairing/unpairing is supported.
16. Assignment conflicts are detected.
17. POS-to-terminal connectivity can be monitored.
18. Terminal-to-payment-service connectivity can be monitored.
19. Payment-server/provider connectivity can be monitored.
20. Card-present transactions can be monitored in real time.
21. Transaction states are clearly represented.
22. Sensitive authentication/card data is not exposed through operational screens.
23. Unknown/uncertain payment states can be identified.
24. POS is prevented from blindly retrying uncertain payments.
25. Idempotency supports duplicate-payment prevention.
26. Reversal status can be represented.
27. Degraded/offline capability is configurable only where supported.
28. Offline transaction limits can be configured.
29. Offline cumulative exposure can be controlled.
30. Stored/offline transactions can be queued for synchronization.
31. Offline transactions can be tracked until synchronization.
32. Terminal health can be monitored.
33. Terminal incidents can be managed.
34. Firmware/software versions can be tracked.
35. Terminal maintenance can be scheduled.
36. Terminal configurations can be simulated/tested before activation.
37. Certification checklists can be completed before production activation.
38. AI can identify terminal operational anomalies without autonomously changing sensitive payment settings.
39. Card-present transaction metadata is available for reconciliation.
40. All terminal configuration respects TICVAI multi-tenancy, RBAC/PBAC, approval and audit requirements.

Board 3 — Final 10 Screens

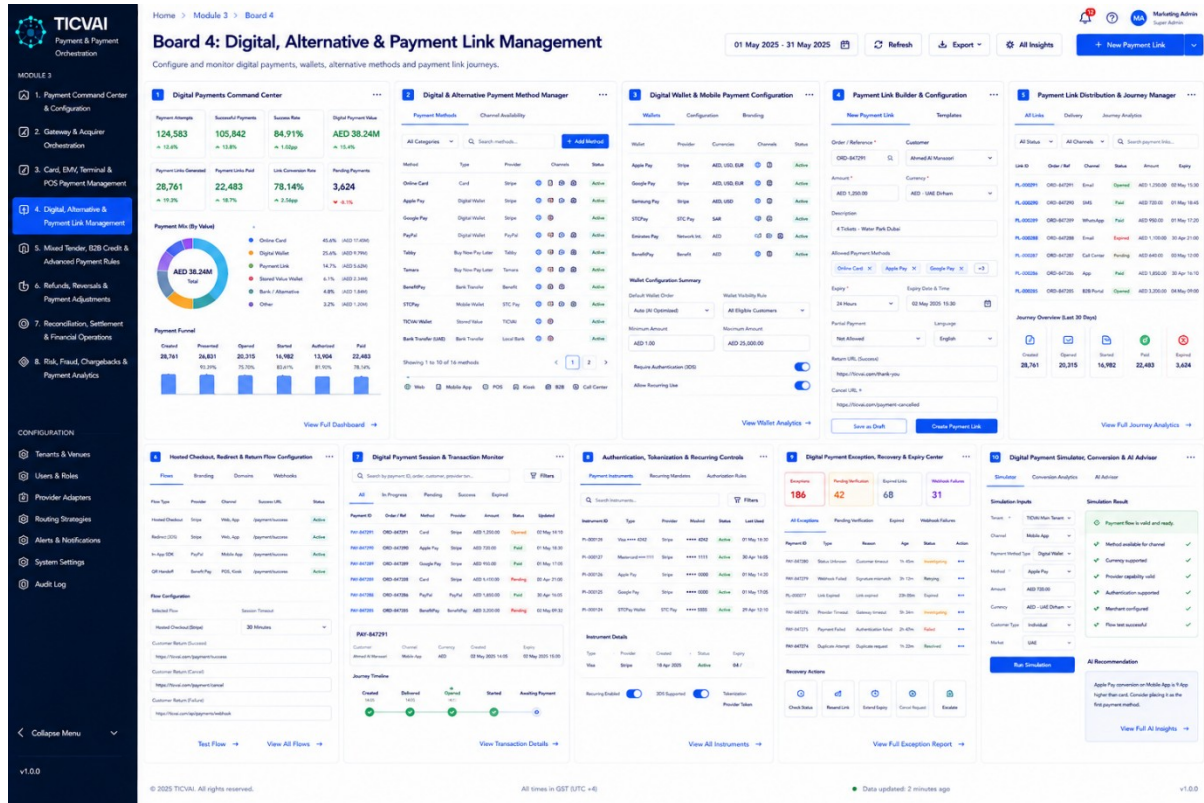
1. **Terminal & Card-Present Command Center**
2. **Payment Terminal & Device Inventory**
3. **Terminal Provisioning & Device Configuration**

- 4. POS, Kiosk & Terminal Assignment Manager**
5. EMV & Card-Present Processing Configuration
- 6. Payment Server & Terminal Connectivity Manager**
7. Card-Present Transaction Monitor & Operations
- 8. Degraded, Offline & Store-and-Forward Manager**
9. Terminal Health, Maintenance & Incident Center
- 10. Terminal Simulator, Certification & AI Operations Advisor**

Board 3 is locked as: *Card, EMV, Terminal & POS Payment Management — 10 backend screens.*

Module 3 — Payment & Payment Orchestration

Board 4 — Digital, Alternative & Payment Link Management



Board Objective

Board 4 shall provide TICVAI with the centralized configuration and operational layer for **digital payments, alternative payment methods, digital wallets, payment links, remote payment requests, and asynchronous payment journeys**.

The responsibility split remains:

- **Board 1:** Which payment methods are allowed?
- **Board 2:** Which gateway/PSP/acquirer processes them?
- **Board 3:** How card-present terminal payments operate.
- **Board 4:** How **card-not-present digital and remote payment experiences** are configured, generated, monitored and completed.

This board is particularly important for:

- B2C checkout
- Mobile App
- Call Center
- B2B
- POS remote collection
- Customer-service payment requests
- Membership/subscription journeys
- Payment links
- Digital wallets
- Partner/API journeys

Board 4 shall contain 10 backend screens.

Screen 1 — Digital Payments Command Center

Purpose

Provide centralized operational visibility across all digital and alternative payment journeys.

KPI Cards

- Digital Payment Attempts
- Successful Payments
- Digital Payment Value
- Success Rate
- Payment Links Generated
- Payment Links Paid
- Payment Link Conversion
- Wallet Transactions
- Alternative Payment Transactions
- Pending Payments
- Expired Payment Requests
- Failed Digital Payments

Payment Mix

Show by:

- Online Card
- Digital Wallet
- Payment Link
- Stored Value
- Bank/Alternative Method
- Other configured digital methods

Channel Performance

Compare:

- B2C
- Mobile App
- Call Center
- B2B
- POS Payment Link
- Partner/API

Status Funnel

Created → Presented/Sent → Opened → Payment Started → Authorized → Paid

with abandonment and failure visibility.

Screen 2 — Digital & Alternative Payment Method Manager

Purpose

Configure the digital payment experiences available through TICVAI.

Method Categories

Support provider-enabled methods such as:

- Online card
- Mobile/digital wallets
- Redirect payments
- Bank-based methods
- Account-to-account methods
- QR-based digital payment where applicable
- Stored-value wallet
- Provider-specific alternative methods
- Future payment methods through adapters

Method Record

For each method:

- Method ID
- Name
- Type
- Provider
- Markets
- Currencies
- Channels
- Customer-facing label
- **Status**
- Minimum amount
- Maximum amount
- Refund capability
- Recurring capability

Availability

Methods should be configurable without modifying the checkout application itself.

Screen 3 — Digital Wallet & Mobile Payment Configuration

Purpose

Manage wallet-based and mobile digital payment methods.

Wallet Configuration

For each supported wallet:

- Wallet name
- Provider
- Gateway
- Merchant account
- Supported market

- Currency
- B2C enabled
- App enabled
- Kiosk enabled where supported
- POS enabled where supported
- Recurring capability
- Refund capability

- **Status**

Examples

The architecture should accommodate provider-supported wallets such as:

- Apple Pay
- Google Pay
- Regional wallets
- Client-specific wallets

without hard-coding TICVAI to a particular brand.

Validation

Check:

- Provider supports wallet
- Merchant account configured
- Domain/app registration where required
- Currency supported
- Channel compatible
- Required certificates/configuration valid

Screen 4 — Payment Link Builder & Configuration

Purpose

Allow authorized users and systems to generate secure payment requests without requiring the customer to be physically present at a POS.

Payment Link Creation

Configure:

- Order / transaction reference
- Customer reference
- Amount
- Currency
- Description
- Payment methods
- Expiry
- Partial payment policy
- Language
- Return destination

- Business unit
- Venue
- Channel/source

Example**Order**

ORD-847291

Amount

AED 1,250

Expiry

24 hours

Allowed

Card + Digital Wallet

Link Status

- Draft
- Active
- Opened
- Payment Started
- Paid
- Expired
- Cancelled
- Failed

Security

Use non-predictable payment references/tokens.

Do not expose sensitive internal information in the URL.

Screen 5 — Payment Link Distribution & Customer Journey Manager

Purpose

Manage how a generated payment request is delivered and how its lifecycle progresses.

Supported Delivery Integrations

Where TICVAI's communications layer supports them:

- Email
- SMS
- Mobile App
- Customer Portal
- B2B Portal
- QR display
- Agent copy/share action
- API

Important Boundary

Board 4 determines:

What payment request should be delivered and its payment state.

The communications platform determines:

How the message is actually sent.

Journey

Created

↓

Delivered

↓

Opened

↓

Payment Started

↓

Payment Completed

or

Expired / Failed / Cancelled

Delivery History

Record:

- Channel
- Destination reference where permitted
- Sent timestamp
- Delivery result
- Opened timestamp
- Payment result

Resend Controls

Authorized users can:

- Resend
- Change approved delivery channel
- Extend expiry
- Cancel

subject to permissions and policy.

Screen 6 — Hosted Checkout, Redirect & Return Flow Configuration

Purpose

Configure digital payment flows that require provider-hosted payment pages, redirects, SDKs, or asynchronous customer journeys.

Flow Types

Support:

- Embedded/hosted component
- Hosted checkout

- Redirect
- App SDK
- QR handoff where applicable
- External payment authorization

Flow Configuration

Configure:

- Provider
- Success return
- Failure return
- Cancel return
- Timeout
- Session expiry
- Callback
- Webhook
- Allowed domains/apps
- Branding profile reference

Example Flow

TICVAI Checkout

↓

Provider Hosted Payment

↓

Customer Authorizes

↓

Provider

↓

Webhook → TICVAI

and

Customer Return → TICVAI

Important

The browser return must **not alone be treated as proof of payment**.

Authoritative payment state should be verified against provider/server-side information.

Screen 7 — Digital Payment Session & Transaction Monitor

Purpose

Provide operational teams with real-time visibility into digital payment sessions.

Search

Search by:

- Payment ID
- Order

- Payment link
- Customer reference
- Provider transaction
- Channel
- Payment method
- Amount
- Currency

- **Status**

- Date/time

Session Details

Show:

- TICVAI Payment ID
- Order ID
- Payment method
- Provider
- Gateway
- Amount
- Currency
- Created
- Session expiry
- Customer journey stage
- Provider status
- TICVAI status

Digital States

Examples:

- Session Created
- Awaiting Customer
- Opened
- Authentication Required
- Processing
- Authorized
- Captured
- Failed
- Declined
- Cancelled
- Expired
- Status Verification Required

Operations

Depending on permissions:

- Check Status

- Cancel Session
 - Regenerate Link
 - View Provider Events
 - View Decision Trace
-

Screen 8 — Authentication, Tokenization & Recurring Payment Controls

Purpose

Manage digital-payment security capabilities and reusable payment references without turning TICVAI into a card vault.

Authentication

Support provider capabilities such as:

- 3-D Secure
- Strong Customer Authentication where applicable
- Provider risk authentication
- Challenge/redirect flows

Tokenization

Store only approved provider/token references required by TICVAI.

Display:

- Payment Instrument Reference
- Provider
- Token Status
- Masked Identifier where allowed
- Card/Instrument Type
- Expiry metadata where provided
- Customer/account relationship
- Created
- Last Used

Token States

- Active
- Expired
- Suspended
- Deleted/Revoked

Recurring Capability

Configure use cases such as:

- Membership
- Subscription
- Installment schedule
- Repeat authorized payment

Consent / Mandate Reference

Where required, retain appropriate reference to:

- Customer authorization
 - Mandate
 - Terms version
 - Authorization timestamp
 - Provider reference
-

Screen 9 — Digital Payment Exception, Recovery & Expiry Center

Purpose

Centralize operational handling of incomplete or abnormal digital-payment journeys.

Exception Types

- Payment Link Expired
- Payment Link Not Delivered
- Customer Abandoned
- Provider Timeout
- Callback Missing
- Webhook Failed
- Status Unknown
- Authentication Failed
- Duplicate Attempt
- Currency Mismatch
- Amount Mismatch
- Payment Completed After UI Timeout

Example

Customer payment page shows:

Timed Out

but provider webhook later reports:

Captured

TICVAI must update the original payment rather than asking the customer to pay again.

Recovery Actions

Depending on condition:

- Query provider
- Reprocess webhook
- Resend payment request
- Extend expiry
- Cancel payment request
- Escalate
- Mark for investigation

Exception SLA

Show:

- Age

- Severity
- Owner
- **Status**
- Resolution

Screen 10 — Digital Payment Simulator, Conversion & AI Advisor

Purpose

Allow administrators to validate digital payment journeys and identify conversion opportunities before deploying configuration changes.

Simulation Inputs

Select:

- Tenant
- Venue
- Channel
- Payment method
- Provider
- Currency
- Amount
- Customer type
- Device context
- Market
- Transaction type

Example Scenario

Channel:

Mobile App

Amount:

AED 720

Currency:

AED

Customer:

Individual

Available Digital Methods

- ✓ Online Card
- ✓ Digital Wallet
- ✓ Payment Link
- × B2B Credit

Flow Simulation

Checkout

↓

Digital Wallet Selected**Provider Capability****Merchant Configuration****Authentication****Authorization****Payment Successful**

Conversion Analytics

Screen 10 should also show digital funnel performance.

Example:

Payment Method Presented

100,000

**Selected**

62,000

**Payment Started**

59,000

**Authenticated**

55,000

**Authorized**

52,000

**Completed**

51,400

Drop-Off Analysis

AI can identify:

6.8% of Mobile App customers abandon during external provider redirect.

or:

Digital Wallet conversion is 9.4 percentage points higher than online card for returning mobile customers.

AI Digital Payment Advisor

AI can recommend:

- Payment method ordering
- Channel-specific methods
- Wallet visibility
- Checkout simplification
- Expiry optimization
- Payment-link timing
- Provider-flow investigation
- Abandonment investigation

Example

Digital Wallet currently appears third on Mobile App checkout but has the highest conversion rate. Consider moving it to the first position.

Another Example

Payment links sent by Call Center have a 28% abandonment rate after 24 hours. Historical data indicates a 6-hour reminder performs better.

AI may create a **draft recommendation**, not directly alter payment configuration.

Payment Link Architecture

A payment link should represent a controlled payment request rather than simply a URL.

Recommended model:

Business Transaction / Order

↓

Payment Request

Contains:

- Payment Request ID
- Amount
- Currency
- Expiry
- Allowed methods
- Customer/order reference
- **Status**

↓

Secure Link Token

↓

Hosted Payment Journey

↓

Payment Transaction

This allows links to be:

- Cancelled
- Expired
- Regenerated
- Audited
- Reconciled

without losing the underlying business transaction.

Payment Link Amount Integrity

This is important.

If a payment request is generated for:

AED 1,250

the customer must not be able to manipulate the browser/link and change the amount.

The authoritative amount must come from TICVAI's server-side payment request.

Order Lock / Price Change Handling

Consider:

Order:

AED 720

↓

Payment Link generated.

↓

Before customer pays, order changes to:

AED 780.

TICVAI must have a configurable policy:

Option A — Invalidate Old Link

Create a new payment request.

Option B — Lock Order

Prevent commercial changes while an active payment request exists.

Option C — Recalculate

Only if explicitly supported by the workflow.

The system should never silently accept an incorrect amount.

Payment Link Expiry

Expiry should be configurable based on use case.

Examples:

Ticket Reservation

15 minutes.

Call Center Booking

2 hours.

B2B Invoice Collection

3 days.

Membership Renewal

7 days.

Expiration should coordinate with:

- Inventory holds
- Seat reservations
- Timeslot capacity
- Order state

This integration is important for ticketing.

Reservation Relationship

Example:

Customer reserves:

4 tickets

↓

Capacity held for:

15 minutes

↓

Payment Link expires in:

15 minutes

If payment is not completed:

Release inventory/capacity

according to the Ticketing/Capacity policy.

Board 4 provides the payment event.

Ticketing/Capacity remains authoritative for the inventory release.

Asynchronous Payment Principle

Some digital payment methods do not complete instantly.

Therefore TICVAI must support:

Pending

without assuming:

Pending = Failed.

Example:

Payment initiated.

↓

Customer authorizes externally.

↓

Provider processes.

↓

TICVAI receives webhook.

↓

Payment becomes:

Captured

The order workflow must support this asynchronous behavior.

Status Authority

I recommend the following priority principle:

Customer Browser

User experience only.

TICVAI Payment Service

Central transaction state.

Provider Server Response / Verified Event

Authoritative processing evidence.

Reconciliation

Final financial verification.

This prevents browser manipulation or connection failures from corrupting transaction status.

Digital Wallet vs TICVAI Wallet

These should be clearly separated.

External Digital Wallet

Examples include device/provider-based payment wallets.

This is a **payment instrument**.

TICVAI Stored-Value Wallet

A customer balance managed by TICVAI.

This is a **stored-value/business capability**.

Both may appear as payment methods, but their accounting and lifecycle are different.

Payment Links from POS

Board 4 should allow POS agents to initiate a remote payment request.

Example:

Customer cannot physically present card.

POS Agent:

Send Payment Link

↓

Customer pays on mobile.

↓

POS receives:

Payment Completed

↓

Order completes.

This is especially useful for:

- Call center
 - Customer service
 - Group bookings
 - B2B
 - Remote sales
-

Recurring Payments

Board 4 should configure the digital-payment capability for recurring transactions.

Example:

Annual Membership

↓

Customer authorizes reusable payment instrument.

↓

Provider token/mandate stored as approved reference.

↓

Future payment request.

The detailed subscription schedule remains with the Membership/Subscription module.

Board 4 owns the **digital payment execution capability**.

Payment Instrument Lifecycle

Reusable instrument references should support:

Created

→ **Active**

→ **Used**

→ **Updated where provider supports**

→ **Expired**

→ **Suspended**

→ **Revoked/Deleted**

Do not assume a token is permanently valid.

Digital Payment Security

Required controls should include:

- TLS-secured communication
- Secure provider adapters
- Webhook signature validation
- Non-predictable payment-link tokens
- Link expiration
- Replay protection

- Idempotency
 - Server-side amount validation
 - Provider status verification
 - Restricted access to payment instrument references
 - Audit logging
-

PCI Boundary

As with Board 3, TICVAI should minimize raw card-data handling.

Preferred online architecture:

Customer Browser / App

↓

Provider-certified Hosted Component / SDK

↓

Provider

↓

Token / Transaction Reference

↓

TICVAI Payment Service

This can materially reduce TICVAI's card-data exposure, although final PCI scope depends on the chosen implementation and providers.

Relationship with Board 1

Board 1 determines:

Digital Wallet is available on Mobile App.

Board 4 manages:

How the Mobile App wallet payment journey operates.

Relationship with Board 2

Board 2 determines:

Which provider processes Digital Wallet.

Board 4 manages:

Customer session, redirect/authentication, callbacks and digital transaction state.

Relationship with Board 5

Board 4 handles a single digital payment journey.

Board 5 will handle more complex tender logic such as:

AED 100 Gift Card + AED 620 Credit Card

or:

B2B Credit + Card

Therefore **mixed tender should not be built into Board 4.**

Relationship with Board 6

Board 4 may show the payment is:

Paid

Board 6 will own the subsequent:

- Refund
- Partial refund
- Void
- Reversal
- Adjustment

workflows.

Relationship with Board 7

Every digital transaction should retain:

- TICVAI Payment ID
- Payment Request ID
- Provider
- Merchant account
- Provider transaction ID
- Amount
- Currency
- Fee references where available
- Payment date/time

Board 7 uses this information for reconciliation.

Recommended Events

Standardized events should include:

- PAYMENT_REQUEST_CREATED
- PAYMENT_LINK_CREATED
- PAYMENT_LINK_DELIVERED
- PAYMENT_LINK_OPENED
- PAYMENT_SESSION_CREATED
- PAYMENT_STARTED
- AUTHENTICATION_REQUIRED
- AUTHENTICATION_COMPLETED
- PAYMENT_AUTHORIZED
- PAYMENT_CAPTURED
- PAYMENT_DECLINED
- PAYMENT_FAILED
- PAYMENT_PENDING
- PAYMENT_EXPIRED

- PAYMENT_CANCELLED
 - WEBHOOK_RECEIVED
 - PAYMENT_STATUS_VERIFIED
-

Roles & Permissions

Recommended roles:

- Payment Administrator
- **E-Commerce Manager**
- Payment Operations
- **Call Center Manager**
- **B2B Manager**
- **Membership Manager**
- Customer Service
- Integration Administrator
- **Finance Manager**
- System Administrator
- Auditor

Sensitive Permissions

Separately control:

- Configure digital payment method
 - Configure digital wallet
 - Create payment link
 - Extend payment link
 - Cancel payment link
 - Access reusable payment references
 - Configure recurring capability
 - Configure authentication
 - Change hosted checkout flow
 - Resolve payment exception
 - Publish digital payment configuration
-

Audit Requirements

Record:

- Payment Request ID
- Payment ID
- User/system
- Tenant
- Channel

- Customer/order reference
- Amount
- Currency
- Creation
- Delivery event
- Expiry
- Changes
- Provider
- Payment outcome
- Cancellation
- Manual intervention

Sensitive credentials or raw payment data must never appear in the audit trail.

Board 4 Acceptance Criteria

Board 4 shall be considered complete when:

1. TICVAI provides a centralized Digital Payments Command Center.
2. Digital payment methods can be configured independently.
3. Alternative payment methods are extensible through provider capabilities.
4. External digital wallets can be configured.
5. External wallets and TICVAI stored-value wallet are distinguished.
6. Digital methods can vary by channel.
7. Digital methods can vary by market and currency.
8. Payment links can be generated.
9. Payment links use secure non-predictable references.
10. Payment-link amount is enforced server-side.
11. Payment links have configurable expiry.
12. Payment links can be cancelled.
13. Payment-link status can be monitored.
14. Payment requests can be delivered through integrated communication channels.
15. Payment-link delivery history can be retained.
16. Payment links can support B2C, POS, Call Center and B2B use cases as permitted.
17. Hosted checkout is supported.
18. Redirect payment journeys are supported.
19. App/SDK payment journeys can be represented.
20. Customer browser return is not treated alone as authoritative payment confirmation.
21. Server-side provider verification is supported.
22. Asynchronous payment states are supported.
23. Pending payments are distinguished from failed payments.
24. Digital payment sessions can be monitored.
25. Authentication/3DS capabilities can be configured.

26. Tokenized/reusable payment references can be managed without exposing raw card credentials.
 27. Recurring-payment capability can be represented.
 28. Appropriate authorization/mandate references can be retained where required.
 29. Payment-link expiry can coordinate with order/inventory reservation policy.
 30. Order changes can invalidate or govern existing payment requests.
 31. Digital payment exceptions are centrally visible.
 32. Failed webhooks/callbacks can be investigated.
 33. Unknown payment states can be verified before another charge is attempted.
 34. Duplicate/replayed requests are protected through idempotency/replay controls.
 35. Digital payment funnel conversion can be measured.
 36. Drop-off can be analyzed by method/channel/provider.
 37. AI can recommend digital-payment optimizations without directly publishing sensitive changes.
 38. Digital transactions retain sufficient information for downstream reconciliation.
 39. Sensitive payment data and credentials are protected.
 40. All digital payment configuration respects TICVAI multi-tenancy, RBAC/PBAC, approval and audit requirements.
-

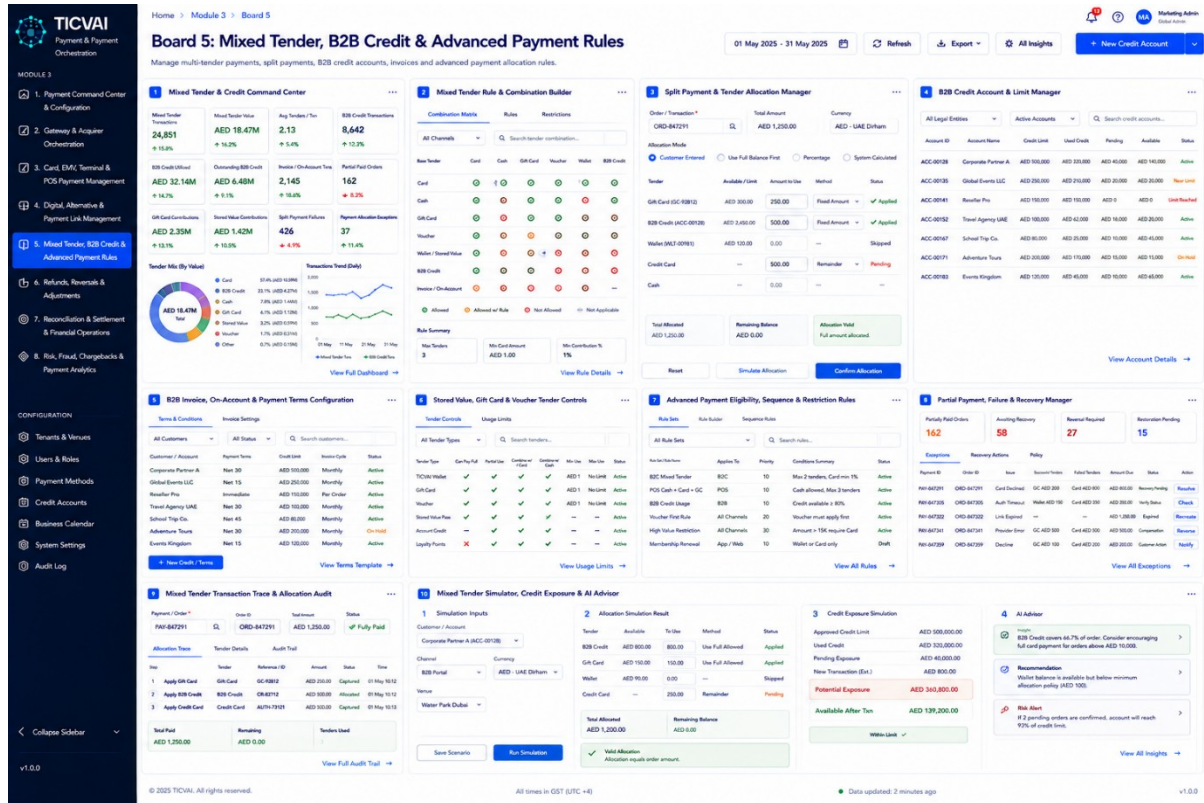
Board 4 — Final 10 Screens

1. **Digital Payments Command Center**
2. **Digital & Alternative Payment Method Manager**
3. **Digital Wallet & Mobile Payment Configuration**
4. **Payment Link Builder & Configuration**
5. **Payment Link Distribution & Customer Journey Manager**
6. **Hosted Checkout, Redirect & Return Flow Configuration**
7. **Digital Payment Session & Transaction Monitor**
8. **Authentication, Tokenization & Recurring Payment Controls**
9. **Digital Payment Exception, Recovery & Expiry Center**
10. **Digital Payment Simulator, Conversion & AI Advisor**

Board 4 is locked as: *Digital, Alternative & Payment Link Management — 10 backend screens.*

Module 3 — Payment & Payment Orchestration

Board 5 — Mixed Tender, B2B Credit & Advanced Payment Rules



Board Objective

Board 5 shall provide TICVAI with the centralized engine for **multi-tender payments, split payments, B2B account credit, invoice/on-account payments, gift card/voucher combinations, payment sequencing, tender restrictions, and advanced payment allocation rules.**

The responsibility split is now:

- **Board 1:** Which payment methods are available?
- **Board 2:** Which gateway/acquirer processes electronic payments?
- **Board 3:** How card-present terminal payments operate.
- **Board 4:** How digital, alternative and payment-link journeys operate.
- **Board 5:** How **multiple payment sources can be combined and allocated to one transaction, including B2B credit and controlled commercial payment arrangements.**

The core question is:

“How can TICVAI safely settle one order using one or several payment sources while ensuring the total paid amount, credit exposure, tender rules and financial allocation remain correct?”

Example:

Order Total: AED 1,000

- Gift Card: AED 200
- Loyalty/Stored Value: AED 100
- Credit Card: AED 700

or:

- B2B Account Credit: AED 800
- Card: AED 200

Board 5 shall contain 10 backend screens.

Screen 1 — Mixed Tender & Credit Command Center

Purpose

Provide centralized operational visibility over mixed-tender and account-credit transactions.

KPI Cards

- Mixed-Tender Transactions
- Mixed-Tender Value
- Average Tenders per Transaction
- B2B Credit Transactions
- B2B Credit Utilized
- Outstanding B2B Credit
- Invoice / On-Account Transactions
- Gift Card Contributions
- Stored-Value Contributions
- Split Payment Failures
- Partially Paid Orders
- Payment Allocation Exceptions

Tender Mix

Show combinations such as:

- Card + Cash
- Gift Card + Card
- Voucher + Card
- Stored Value + Card
- B2B Credit + Card
- Multiple Cards
- Cash + Card + Voucher

Alerts

Examples:

12 partially paid orders require resolution.

Corporate Account ABC has reached 92% of its approved credit limit.

4 mixed-tender transactions have unresolved refund allocation.

Screen 2 — Mixed Tender Rule & Combination Builder

Purpose

Define which payment methods may be combined within a single transaction.

Combination Matrix

Tender	Card	Cash	Gift Card	Voucher	Wallet	B2B Credit
Card	✓	✓	✓	✓	✓	✓

Tender	Card	Cash	Gift Card	Voucher	Wallet	B2B Credit
Cash	✓	—	✓	✓	✓	✓
Gift Card	✓	✓	✓	✓*	✓	✓
Voucher	✓	✓	✓*	—	✓	✓

*Subject to configured business rules.

Combination Rules

Configure:

- Allowed combination
- Prohibited combination
- Maximum number of tenders
- Minimum contribution
- Maximum contribution
- Channel
- Venue
- Product/category
- Customer type
- Currency

Example

B2C

Allow:

Gift Card + Card.

Maximum:

2 tenders.

Another Example

POS

Allow:

Cash + Card + Gift Card.

Maximum:

3 tenders.

Screen 3 — Split Payment & Tender Allocation Manager

Purpose

Determine how an order total is allocated across several payment methods.

Example

Order Total

AED 1,250

Tender Allocation

Gift Card:

AED 250

B2B Credit:

AED 500

Card:

AED 500

Remaining Balance

AED 0

Allocation Modes

Support:

- Customer-entered amount
- Fixed amount
- Percentage
- Use full available balance
- Use maximum allowed amount
- System-calculated remainder

Sequence

Example:

1. Voucher
2. Gift Card
3. Account Credit
4. Card

Validation

System must ensure:

Sum of successful tender allocations = Amount Due

before completing the order.

Screen 4 — B2B Credit Account & Limit Manager

Purpose

Manage payment credit facilities granted to authorized B2B customers, resellers, corporate clients or partners.

This is not the general B2B customer profile.

The B2B/CRM module remains authoritative for the customer/account itself.

Board 5 owns the **payment-credit capability**.

Credit Account

Display:

- B2B Account
- Credit Account ID
- Legal Entity
- Currency
- Approved Credit Limit
- Used Credit

- Available Credit
- Pending Exposure
- Overdue Exposure
- **Status**

Example

Corporate Partner A

Approved Limit:

AED 500,000

Used:

AED 320,000

Pending:

AED 40,000

Available:

AED 140,000

Credit Status

- Active
- Near Limit
- Limit Reached
- Suspended
- On Hold
- Expired

Controls

Configure:

- Credit limit
- Effective dates
- Currency
- Allowed business units
- Allowed channels
- Payment terms
- Approval level

Screen 5 — B2B Invoice, On-Account & Payment Terms Configuration

Purpose

Configure how approved B2B customers can transact without immediate full payment.

Payment Terms

Support configurable terms such as:

- Immediate
- Net 7
- Net 15
- Net 30

- Net 45
- Net 60
- Custom

Account Payment Methods

Examples:

- B2B Credit
- Invoice
- Corporate Account
- Reseller Account
- On Account

Configuration

Define:

- Customer/account
- Credit agreement
- Payment term
- Currency
- Billing cycle
- Invoice requirement
- Purchase order requirement
- Approval requirement
- Transaction maximum
- Outstanding balance limit

Example

Reseller ABC

Payment Method:

B2B Credit.

Terms:

Net 30.

Credit Limit:

AED 250,000.

PO Required:

Yes.

Important Boundary

The detailed invoice-generation/accounting process should remain integrated with **Finance**.

Board 5 controls whether the transaction may be placed **on account**.

Screen 6 — Stored Value, Gift Card & Voucher Tender Controls

Purpose

Manage how stored-value instruments participate in payment without duplicating the Wallet or Voucher modules.

Sources

Consume balances/validity from authoritative modules such as:

- TICVAI Wallet
- Gift Card
- Stored Value
- Voucher
- Account Credit

Payment Controls

Configure:

- Can be used as tender
- Can combine with card
- Can combine with cash
- Can combine with B2B credit
- Minimum redemption
- Maximum redemption
- Full/partial balance usage
- Currency restrictions
- Product restrictions
- Venue restrictions
- Channel restrictions

Example

Gift Card Balance:

AED 300

Order:

AED 720

Use:

AED 300 Gift Card

-

AED 420 Card.

Important

Board 5 should **consume the available balance** from the Wallet/Gift Card service.

It should not maintain a second independent stored-value balance.

Screen 7 — Advanced Payment Eligibility, Sequence & Restriction Rules

Purpose

Provide advanced transaction-level payment rules beyond the general method availability configured in Board 1.

Board 1 answers:

Is this payment method available?

Board 5 answers:

Can it be combined with these other tenders, in this order, for this amount and transaction?

Rule Inputs

- Channel
- Venue
- Product
- Product category
- Basket amount
- Customer type
- B2B account
- Membership
- Currency
- Payment method
- Tender combination
- Remaining balance

Example

IF

Customer = Approved B2B

AND

Available Credit $\geq$ 80% of Order

THEN

Allow:

B2B Credit + Card.

Sequence Rules

Example:

Voucher first

↓

Gift Card second

↓

B2B Credit third

↓

Card for remainder

Restrictions

Support:

- Maximum tenders
- Minimum card amount
- Cash restriction
- Credit restriction
- Voucher exclusivity
- Stored-value limit
- Tender priority

Screen 8 — Partial Payment, Failure & Recovery Manager

Purpose

Handle situations where some tenders succeed but another tender fails.

This is one of the most important screens in Board 5.

Example

Order:

AED 1,000

Gift Card:

AED 200 ✓

Card:

AED 800 ✕

The system now has:

AED 200 successfully consumed

but:

AED 800 outstanding.

The order must not simply be treated as fully failed without understanding the successful first tender.

Recovery Options

Depending on policy:

- Retry failed tender
- Select another tender
- Pay remaining balance
- Reverse successful tender
- Restore gift card balance
- Cancel transaction
- Hold order pending payment
- Escalate

Partial Payment States

- Payment Started
- Partially Paid
- Awaiting Remaining Tender
- Recovery Required
- Reversal Pending
- Fully Paid
- Cancelled

Timeout

Configure how long an order may remain:

Partially Paid

before recovery action is required.

Screen 9 — Mixed Tender Transaction Trace & Allocation Audit**Purpose**

Provide a complete financial and operational explanation of every multi-tender transaction.

Example**Order ORD-847291**

Total:

AED 1,000

Tender 1

Gift Card

Amount:

AED 200

Status:

Captured

Reference:

GC-92812

Tender 2

B2B Credit

Amount:

AED 300

Status:

Allocated

Reference:

CR-82712

Tender 3

Credit Card

Amount:

AED 500

Status:

Captured

Provider:

Provider A

Order Payment State**Fully Paid****Trace****Order Total**

↓

Eligible Tenders

↓

Combination Rule

↓

Allocation

↓

Tender 1

↓

Tender 2

↓

Tender 3

↓

Total Validation

↓

Paid**Audit Information**

Capture:

- Operator/customer
- Channel
- Rules applied
- Payment IDs
- Tender references
- Amounts
- Currency
- Timestamps
- Overrides
- Approval
- Reversal/refund relationship

Screen 10 — Mixed Tender Simulator, Credit Exposure & AI Advisor**Purpose**

Allow administrators to test complex tender combinations before activating rules.

Simulation Inputs

Select:

- Tenant
- Venue
- Channel
- Customer
- B2B account
- Order value
- Currency
- Available gift-card balance
- Available wallet balance

- Available B2B credit
- Selected payment methods

Example

Order:

AED 1,200

Customer:

Corporate Account

Available B2B Credit:

AED 800

Gift Card:

AED 150

Card:

Available

Simulation Result

Recommended allocation:

B2B Credit:

AED 800

Gift Card:

AED 150

Card:

AED 250

Total:

AED 1,200

Validation:

✓

Credit Exposure Simulation

Example:

Current Credit Limit:

AED 500,000

Current Used:

AED 430,000

Pending Orders:

AED 35,000

New Transaction:

AED 50,000

Potential Exposure:

AED 515,000

Result:

Credit Limit Exceeded

Available actions according to policy:

- Require card for excess
- Request approval
- Reject B2B credit
- Reduce credit allocation

AI Advisor

AI can provide recommendations such as:

Corporate Account A is expected to exceed its approved credit limit within five days based on current bookings and pending orders.

31% of failed mixed-tender transactions occur after stored value is consumed and the final card payment fails. Consider card authorization before final stored-value commitment where technically and commercially appropriate.

Allowing Gift Card + Digital Wallet on Mobile App is estimated to recover AED 180K in otherwise abandoned transactions.

AI should create:

- Insight
- Simulation
- Draft rule
- Risk alert

but should not autonomously change credit limits or financial authorization.

Critical Transaction Principle — Atomicity

Mixed tender introduces a major technical challenge:

What happens when one payment succeeds and another fails?

For example:

AED 1,000 order

↓

Gift Card AED 200

↓

Card AED 800

The platform should not leave the customer with:

- AED 200 deducted
- No ticket/order
- No clear recovery process

Therefore every mixed-tender transaction needs an orchestrated transaction state.

Recommended Mixed Tender Flow

Order Total

AED 1,000

↓

Validate Tender Combination

↓

Validate Balances / Credit

↓

Reserve / Prepare Tenders Where Supported

↓

Process Tenders

↓

Validate Total Paid

↓

If successful:

Commit Order

If failure:

Recovery / Compensation Flow

This is effectively a controlled financial orchestration process.

Reservation vs Commitment

Where supported by the underlying instrument, TICVAI should distinguish:

Reserve

Temporarily secure value.

Commit

Permanently consume value.

Example:

Gift Card:

Reserve AED 200.

↓

Card authorization:

AED 800.

↓

Card successful.

↓

Commit gift card AED 200.

This reduces partial-payment problems.

However, not every provider or stored-value system will support reservation.

The architecture therefore needs **compensating transactions** when true atomic execution is impossible.

Compensating Transaction

Example:

Gift Card consumed:

AED 200.

Card fails:

AED 800.

If order is cancelled:

TICVAI should create:

Gift Card Restore +AED 200

rather than manually changing the balance.

This maintains an auditable financial trail.

Multiple Card Payments

Board 5 should support configurable multi-card payments where the relevant channel/provider supports them.

Example:

Order:

AED 1,000

Card 1:

AED 600

Card 2:

AED 400

Controls

- Maximum cards
 - Minimum card amount
 - Channel
 - Customer/operator permission
 - Timeout
 - Failure handling
-

Cash + Card

POS example:

Order:

AED 720

Customer pays:

Cash AED 200

Card AED 520

TICVAI must track both tenders separately while maintaining one order payment status.

Overpayment

The engine must prevent unintended:

Paid Amount > Amount Due

unless a specific business process permits it.

Example:

Amount Due:

AED 720

Tender 1:

AED 300

Tender 2 request:

AED 500

Result:

Maximum Remaining Amount

AED 420

The second tender should not accidentally process AED 500.

Underpayment

If:

Amount Due:

AED 720

Paid:

AED 700

Order status:

Partially Paid

not:

Paid

The system should clearly display:

Remaining Balance: AED 20

Currency Principle

A single mixed-tender transaction should normally have a defined transaction/accounting currency even when provider settlement currencies differ.

Example:

Order:

AED 1,000.

Gift Card:

AED 200.

Card:

AED 800.

Provider later settles card differently according to gateway configuration.

Board 5 manages the **order-level allocation**.

Board 7 manages settlement/reconciliation differences.

B2B Credit Exposure

Credit availability should consider more than invoiced balance.

Recommended formula conceptually:

Approved Credit Limit

– **Outstanding Posted Credit**
– **Pending / Reserved Exposure**
= **Available Credit**

This prevents two simultaneous large B2B transactions from both consuming the same remaining credit.

Credit Reservation

Example:

Available Credit:

AED 100,000.

Transaction A begins:

AED 80,000.

Transaction B begins simultaneously:

AED 70,000.

Without reservation, both might initially appear eligible.

TICVAI therefore needs an atomic/reservation mechanism to prevent over-exposure.

Credit Override

Authorized roles may override certain credit restrictions where the client's policy allows it.

Example:

Credit limit:

AED 100,000.

Transaction causes:

AED 105,000 exposure.

Authorized manager may:

Approve Temporary AED 5,000 Override

Audit must capture:

- User
- Reason
- Amount
- Approval
- Expiry

Do not permanently increase the customer's credit limit through a transaction override.

Promotion & Payment Interaction

The order total should already reflect:

- Pricing
- Tax
- Promotions
- Discounts
- Fees

before final tender allocation.

Conceptually:

Gross Price

↓

Promotion

↓

Tax / Fees

↓

Amount Due

↓

Payment Allocation

Payment should not recalculate the promotion independently.

Payment-Method Promotions

There is one special interaction.

Example:

10% discount when paying with Bank Card X.

The Promotion Engine determines:

Benefit eligibility

based on payment method.

The Payment Engine provides:

selected tender/payment context.

If the customer changes tender:

Bank Card X

→ Cash

the Promotion Engine must revalidate the order.

This prevents retaining a payment-method discount after switching payment method.

Refund Implications

Mixed tender makes refund allocation more complex.

Example:

Original:

Gift Card AED 200

Card AED 800.

Refund:

AED 300.

Possible policy:

1. Return AED 200 to Gift Card.
2. Return AED 100 to Card.

or another governed allocation rule.

Board 5 should preserve original allocation.

Board 6 will own the actual refund allocation and execution rules.

Relationship with Board 1

Board 1 determines:

Gift Card and Card are available.

Board 5 determines:

Gift Card + Card may be combined.

Relationship with Board 2

Board 5 determines:

Card remainder:

AED 500

Board 2 determines:

Which provider processes that AED 500.

Relationship with Board 3

For POS:

Board 5 determines:

Cash AED 200 + Card AED 520.

Board 3 processes the card-present AED 520 through the assigned terminal.

Relationship with Board 4

For Mobile App:

Board 5 determines:

Gift Card AED 200 + Digital Wallet AED 520.

Board 4 manages the digital-wallet payment journey.

Relationship with Finance

Finance remains authoritative for:

- Accounts receivable
- General ledger
- Invoice accounting
- Settlement accounting
- Financial postings

Board 5 manages the **payment orchestration and credit authorization required at transaction time.**

Relationship with B2B / CRM

B2B/CRM owns:

- Company
-

- Reseller
- Partner
- Contact
- Contractual relationship

Board 5 owns:

- Credit limit
- Payment terms
- Credit exposure
- Tender eligibility
- Transaction-level credit authorization

This avoids duplication.

Standardized Events

Recommended events:

- MIXED_TENDER_STARTED
 - TENDER_ADDED
 - TENDER_REMOVED
 - TENDER_RESERVED
 - TENDER_AUTHORIZED
 - TENDER_COMMITTED
 - TENDER_FAILED
 - PARTIAL_PAYMENT_CREATED
 - REMAINING_BALANCE_UPDATED
 - CREDIT_RESERVED
 - CREDIT_RELEASED
 - CREDIT_CONSUMED
 - CREDIT_LIMIT_EXCEEDED
 - PAYMENT_FULLY_ALLOCATED
 - COMPENSATING_TRANSACTION_STARTED
 - COMPENSATING_TRANSACTION_COMPLETED
-

Roles & Permissions

Recommended roles:

- Payment Administrator
- **Finance Manager**
- **B2B Manager**
- Credit Controller

- **Commercial Manager**
- **POS Manager**
- Customer Service
- **Revenue Manager**
- System Administrator
- Auditor

Sensitive Permissions

Separately control:

- Configure tender combinations
- Change tender sequence
- Configure B2B credit
- Change credit limit
- Change payment terms
- Override credit restriction
- Configure partial-payment recovery
- Restore/compensate stored value
- Approve advanced payment rule
- View credit exposure

Audit Requirements

Every multi-tender transaction must retain:

- Order
- Tender
- Payment ID
- Tender type
- Amount
- Currency
- Sequence
- **Status**
- Provider/reference
- Balance before/after where appropriate
- Credit reservation
- User/operator
- Override
- Approval
- Timestamp

Never collapse several tenders into one generic "Paid AED 1,000" record.

Board 5 Acceptance Criteria

Board 5 shall be considered complete when:

1. TICVAI provides a centralized Mixed Tender & Credit Command Center.
2. Multiple payment methods can be used for one order.
3. Allowed tender combinations are configurable.
4. Prohibited combinations are configurable.
5. Maximum number of tenders is configurable.
6. Card + cash can be supported where applicable.
7. Gift card + card can be supported.
8. Voucher + another tender can be supported according to policy.
9. Stored value + another tender can be supported.
10. B2B credit + another tender can be supported.
11. Multiple-card payments can be supported where the channel/provider permits.
12. Tender amounts can be allocated explicitly.
13. Remaining balance is calculated in real time.
14. Overpayment is prevented unless explicitly supported.
15. Underpayment produces a partial-payment state.
16. Tender sequence can be configured.
17. B2B credit limits are supported.
18. Available B2B credit considers pending exposure.
19. Credit reservation prevents concurrent overuse.
20. Payment terms can be configured.
21. Invoice/on-account eligibility can be represented.
22. Credit overrides can require authorization.
23. Transaction overrides do not silently change permanent credit limits.
24. Wallet/gift-card balances remain authoritative in their source modules.
25. Mixed-tender orchestration consumes rather than duplicates stored-value balances.
26. Partial-payment failures are recoverable.
27. Successful earlier tenders are not silently lost when a later tender fails.
28. Compensating transactions can restore consumed value where required.
29. Mixed-tender transactions maintain a complete allocation trace.
30. Every tender retains its own payment/reference status.
31. The total successful allocation must equal amount due before normal order completion.
32. Payment-method promotions can be revalidated when the tender changes.
33. Original tender allocation is retained for future refund processing.
34. Administrators can simulate tender combinations.
35. Credit exposure can be simulated before activation.
36. Invalid combinations are explained.
37. AI can identify payment and credit risks without autonomously changing financial limits.
38. Mixed-tender data is available for downstream reconciliation.

39. All overrides and compensation activities are audited.
 40. All configuration respects TICVAI multi-tenancy, RBAC/PBAC, approval and financial governance requirements.
-

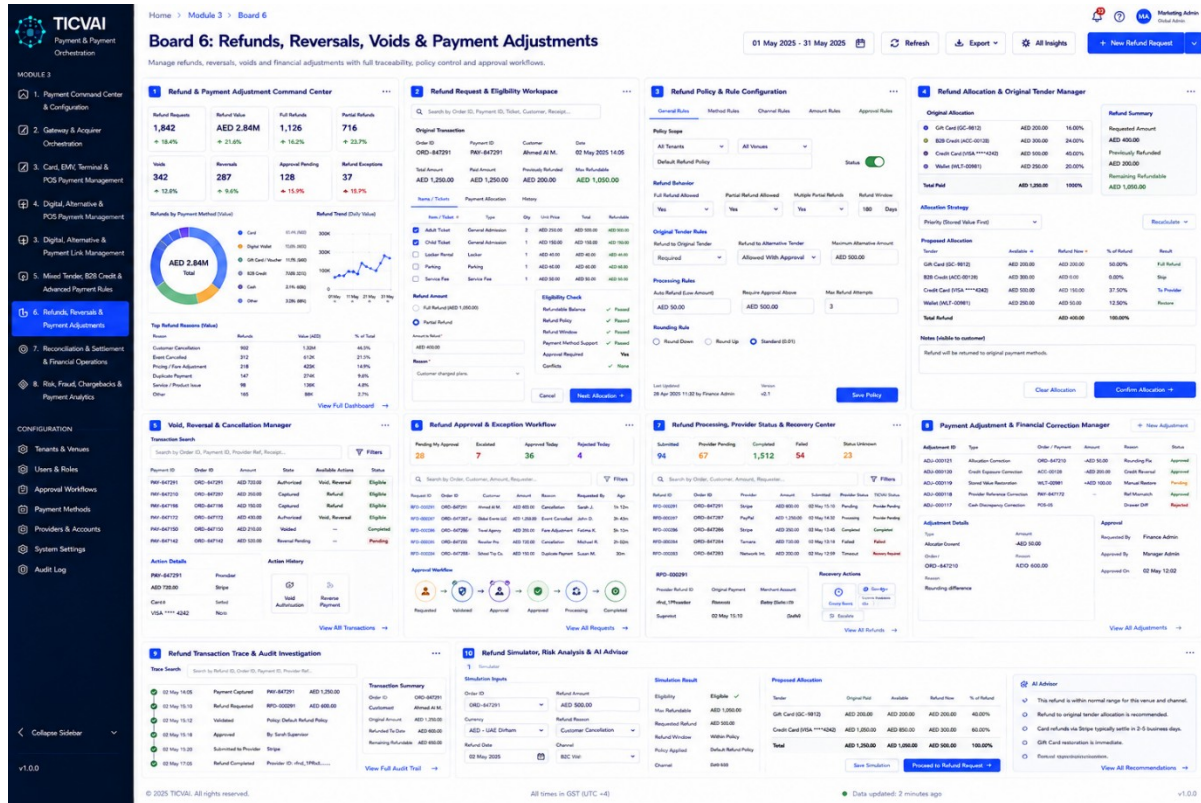
Board 5 — Final 10 Screens

1. **Mixed Tender & Credit Command Center**
2. **Mixed Tender Rule & Combination Builder**
3. **Split Payment & Tender Allocation Manager**
4. **B2B Credit Account & Limit Manager**
5. **B2B Invoice, On-Account & Payment Terms Configuration**
6. **Stored Value, Gift Card & Voucher Tender Controls**
7. **Advanced Payment Eligibility, Sequence & Restriction Rules**
8. **Partial Payment, Failure & Recovery Manager**
9. **Mixed Tender Transaction Trace & Allocation Audit**
10. **Mixed Tender Simulator, Credit Exposure & AI Advisor**

Board 5 is locked as: *Mixed Tender, B2B Credit & Advanced Payment Rules — 10 backend screens.*

Module 3 — Payment & Payment Orchestration

Board 6 — Refunds, Reversals, Voids & Payment Adjustments



Board Objective

Board 6 shall provide TICVAI with the centralized engine for managing the **financial return or correction of previously processed payments**, including:

- Full refunds
- Partial refunds
- Multi-item refunds
- Mixed-tender refund allocation
- Voids
- Payment reversals
- Failed and pending refunds
- Payment corrections/adjustments
- Approval workflows
- Original-tender rules
- Exception recovery
- Complete financial auditability

The responsibility split remains:

Board 1 — Payment methods and configuration

Board 2 — Gateway/acquirer orchestration

Board 3 — Card-present and terminal processing

Board 4 — Digital and payment-link processing

Board 5 — Mixed tender, B2B credit and payment allocation

Board 6 — Returning, cancelling, reversing or correcting money already processed

The key question is:

“When a payment must be cancelled, reversed, partially or fully returned, how does TICVAI determine the correct amount, original tender, provider, approval, financial treatment and resulting payment state?”

Board 6 shall contain 10 backend screens.

Screen 1 — Refund & Payment Adjustment Command Center

Purpose

Provide centralized visibility across refund, reversal, void and adjustment operations.

KPI Cards

- Refund Requests
- Refund Value
- Full Refunds
- Partial Refunds
- Refund Rate
- Pending Refunds
- Failed Refunds
- Voids
- Reversals
- Manual Adjustments
- Approval Pending
- Refund Exceptions

Refund Funnel

Requested

↓

Validated

↓

Approved

↓

Submitted

↓

Provider Processing

↓

Completed

or:

Failed / Exception

Analysis

Break down by:

- Venue
- Channel

- Payment method
- Provider
- Product
- Refund reason
- Operator
- Customer segment
- Full vs partial
- Automatic vs manual

Alerts

Examples:

18 refunds have remained in Provider Pending status for more than 24 hours.

Provider B refund failure rate increased from 1.2% to 6.8%.

Mixed-tender refund exceptions require allocation review.

Screen 2 — Refund Request & Eligibility Workspace

Purpose

Provide the operational workspace for creating and validating a refund request.

Search

Find original transaction using:

- Order ID
- Payment ID
- Ticket
- Customer reference
- Receipt
- Provider transaction
- POS
- Date
- Amount

Original Transaction

Display:

Order: ORD-847291

Original Amount: AED 1,250

Paid: AED 1,250

Previously Refunded: AED 200

Maximum Remaining Refundable: AED 1,050

Refund Selection

Allow:

- Entire transaction
- Selected items
- Selected tickets
- Selected quantities

- Specific monetary amount
- Eligible fees where permitted

Eligibility Checks

Before allowing the refund:

- Transaction exists
- Payment was successful
- Refundable balance remains
- Ticket/product refund policy permits it
- Refund window is valid
- Payment method supports refund
- Provider supports refund
- Required approval obtained
- No conflicting operation exists

Result

Eligible for Refund

or:

Refund Restricted

with exact reason.

Screen 3 — Refund Policy & Rule Configuration

Purpose

Define the **payment execution rules** governing refunds without duplicating Ticket Service Policy.

This boundary is important.

The Ticketing policy determines:

Is the commercial transaction refundable and how much should be refunded?

Board 6 determines:

How should the approved financial refund be executed?

Configuration

Rules by:

- Payment method
- Provider
- Channel
- Venue
- Currency
- Transaction type
- Amount
- Refund age
- Original tender
- Customer/account type

Rules

Configure:

- Full refund allowed
- Partial refund allowed
- Multiple partial refunds allowed
- Refund to original tender required
- Alternative tender permitted
- Maximum refund amount
- Refund approval threshold
- Refund expiry/window
- Automatic vs manual processing

Example

Online Card

Refund to original card:

Required.

Partial Refund:

Allowed.

Alternative Cash Refund:

Not allowed.

Screen 4 — Refund Allocation & Original Tender Manager

Purpose

Determine where the refunded value must be returned.

This screen becomes particularly important for Board 5 mixed-tender transactions.

Example

Original order:

AED 1,000

Paid by:

- Gift Card — AED 200
- B2B Credit — AED 300
- Card — AED 500

Refund requested:

AED 400

Allocation Policy Example

1. Gift Card — AED 200
2. Card — AED 200

or another approved tenant policy.

Allocation Strategies

Support configurable approaches such as:

- Original tender proportion
- Original tender priority

- Reverse tender sequence
- Exact item/tender relationship where available
- Restore stored value first
- Provider-required original tender
- Manual allocation with approval

Critical Rule

Refund allocation must never exceed:

Original captured/consumed amount – previous refunds
for each tender.

Screen 5 — Void, Reversal & Cancellation Manager

Purpose

Clearly distinguish **voids and reversals from refunds**.

These should not all be called “refund”.

Void

Typically used when a transaction can still be cancelled before financial completion/settlement according to provider capability.

Example:

Card payment authorized.

Order cancelled immediately.

Action:

Void Authorization

Reversal

Used to reverse an authorization/payment state, commonly following:

- Communication failure
- Order completion failure
- Terminal timeout
- Provider uncertainty
- Technical exception

Refund

Used when captured/settled value needs to be returned.

Decision Workspace

Display:

Payment State

Captured

Available Actions

Refund ✓

Void ✕

Reversal ✕

Another Example

Payment State:

Authorized

Available:

Void ✓

Refund ✕

Reversal ✓ where supported

Important

Available actions must be derived from actual provider capability and transaction state.

Screen 6 — Refund Approval & Exception Workflow

Purpose

Provide governed approval for financially sensitive refund actions.

Approval Rules

Approval can depend on:

- Refund amount
- Percentage of original transaction
- Refund reason
- Channel
- Product
- Operator role
- Customer type
- Manual override
- Alternative tender
- Refund age

Example

Refund ≤ AED 500

→ Supervisor.

AED 501–5,000

→ Venue Manager.

AED 5,000

→ Finance Manager + Commercial Manager.

Workflow

Requested

↓

Policy Validated

↓

Approval Required

↓

Approver

↓

Approved / Rejected**Execution****Approval Inbox**

Show:

- Request
- Order
- Customer
- Amount
- Reason
- Original payment
- Requested refund method
- Risk indicator
- Requester
- Age

Segregation of Duties

TICVAI should support:

Requester cannot approve their own refund

where tenant policy requires it.

Screen 7 — Refund Processing, Provider Status & Recovery Center**Purpose**

Track refund execution through the appropriate provider and recover failed or uncertain operations.

Refund Processing States

- Created
- Validation Pending
- Approval Pending
- Approved
- Submitted
- Provider Pending
- Completed
- Partially Completed
- Failed
- Status Unknown
- Recovery Required
- Cancelled

Provider Information

Show:

- Provider
- Merchant account

- Original provider transaction
- Refund transaction ID
- Refund amount
- Currency
- Submitted timestamp
- Provider status
- Provider response
- Expected completion

Failure Example

Refund:

AED 500

Provider:

Provider A

Result:

Timeout / Status Unknown

Action:

Verify Provider Status

Do not blindly send another refund request.

Recovery Actions

- Query Provider
- Retry where safe
- Reprocess callback
- Resubmit where confirmed not processed
- Escalate
- Manual review

Screen 8 — Payment Adjustment & Financial Correction Manager**Purpose**

Handle governed payment corrections that are not normal customer refunds.

This must be tightly controlled.

Adjustment Types

Examples:

- Payment allocation correction
- Tender classification correction
- Cash discrepancy correction
- B2B credit correction
- Stored-value restoration
- Provider reference correction
- Transaction-state correction following verified evidence

Important Principle

TICVAI should **not allow administrators to casually edit completed payment records.**

Instead:

Original Financial Record

remains immutable.

↓

Adjustment Record

is created.

Example

Incorrect:

Change payment from AED 500 to AED 450.

Correct:

Original Payment:

AED 500.

Adjustment:

—AED 50.

Reason:

Approved correction.

Adjustment Requirements

- Reason
- Supporting reference
- Approval
- Effective date
- User
- Financial posting impact
- Audit trail

Screen 9 — Refund Transaction Trace & Audit Investigation

Purpose

Provide complete traceability from original sale through refund/reversal/adjustment.

Search

By:

- Refund ID
- Payment ID
- Order
- Ticket
- Provider reference
- Customer
- POS
- Operator
- Date

Example Trace**ORD-847291**

Original Sale:

AED 1,000

↓

Payment Allocation:

Gift Card AED 200

Card AED 800

↓

Refund Request:

AED 300

↓

Eligibility:

Passed

↓

Approval:

Supervisor Approved

↓

Allocation:

Gift Card AED 200

Card AED 100

↓

Execution:

Gift Card Restored ✓

Card Provider Refund ✓

↓

Refund Completed**Audit Timeline**

Show:

- Who requested
- Why
- Policy applied
- Calculated amount
- Manual changes
- Approval
- Provider submission
- Provider result
- Wallet restoration
- Finance posting
- Final status

Screen 10 — Refund Simulator, Risk Analysis & AI Advisor

Purpose

Allow administrators to simulate refund outcomes before execution or rule publication.

Simulation Inputs

Select:

- Order
- Payment
- Refund amount
- Selected items
- Refund reason
- Customer
- Channel
- Original tender
- Provider
- Refund date

Example

Original:

AED 1,200

Payment:

Gift Card AED 200

Card AED 1,000

Previously Refunded:

AED 100

Requested:

AED 500

Simulation

Maximum refundable:

AED 1,100 ✓

Ticket policy:

Eligible ✓

Payment provider:

Refund supported ✓

Approval:

Required ✓

Proposed Allocation

Gift Card:

AED 100

Card:

AED 400

Result

Refund Valid — Manager Approval Required

AI Refund Advisor

AI can identify operational and financial anomalies.

Examples:

Refund rate for Operator Group A is 2.7× the venue baseline.

38% of failed refunds this week originate from Provider C.

Refunds for Product X increased 41% after the latest schedule change.

This refund request is unusual because the transaction is 180 days old and the amount represents 100% of the original sale.

AI Actions

AI may:

- Flag risk
- Explain anomaly
- Recommend review
- Suggest refund allocation
- Identify provider issue
- Recommend policy investigation

AI should **not autonomously approve or execute high-risk refunds**.

Refund Amount Architecture

Refund amount should originate from the relevant business policy.

Conceptually:

Original Order

↓

Products / Tickets / Services

↓

Ticket Service Policy / Commercial Policy

↓

Refundable Amount

↓

Payment Board 6

determines:

- Tender allocation
- Provider execution
- Approval
- Financial state

This avoids Board 6 becoming another refund-policy engine for ticket products.

Refund vs Ticket Cancellation

These are related but separate.

Example:

Customer cancels a ticket.

Ticketing

Determines:

Ticket cancellation permitted.

Refundable value:

AED 180.

Payment

Receives:

Refund AED 180

and determines:

Where and how do we return AED 180?

This separation is critical.

Partial Refund Architecture

Example:

Original:

AED 1,000.

First refund:

AED 200.

Second refund:

AED 300.

Remaining refundable:

AED 500.

The system must always calculate:

Original Captured Amount

– **Completed Refunds**

– **Reserved/Pending Refunds**

= **Available Refundable Balance**

Pending refunds should be included to prevent simultaneous over-refunding.

Refund Reservation / Concurrency Control

Example:

Refundable:

AED 500.

Agent A requests:

AED 400.

At the same time Agent B requests:

AED 300.

Without locking/reservation:

Total potential refund:

AED 700.

Therefore TICVAI must reserve refund exposure while a refund is processing.

Mixed-Tender Refund Example

Original transaction:

AED 1,200

- Gift Card AED 200
- Wallet AED 200
- Card AED 800

Refund:

AED 500

Configured rule:

Return to original tender proportionally.

Original ratios:

Gift Card: 16.67%

Wallet: 16.67%

Card: 66.66%

Approximate refund allocation:

Gift Card: AED 83.35

Wallet: AED 83.35

Card: AED 333.30

The system should manage rounding deterministically.

Original Tender Principle

Where possible and required:

Refund to the original payment instrument.

This supports:

- Financial control
- Fraud prevention
- Provider rules
- Auditability

Alternative refund methods should require explicit policy and potentially approval.

Cash Refunds

POS cash refunds require special operational control.

Support:

- Cash drawer/shift availability
- Maximum refund amount
- Supervisor approval

- Original cash transaction verification
- Cash balance impact

Detailed cash-drawer operations remain part of POS/Finance, but Board 6 controls the payment refund action.

B2B Credit Refund

Example:

B2B Credit consumed:

AED 10,000.

Refund:

AED 2,000.

Instead of sending money:

Credit Exposure

should normally be reduced/restored according to policy.

Result:

B2B available credit increases by:

AED 2,000

while Finance handles associated accounting entries.

Gift Card / Wallet Refund

If original payment used stored value:

Refund may restore value to:

- Original wallet
- Original gift card
- Controlled replacement instrument where permitted

Board 6 sends the approved restoration transaction to the authoritative Wallet/Gift Card service.

It must not directly edit the balance.

Refund Fees

If commercial policy allows a cancellation/refund fee:

Example:

Original:

AED 500.

Cancellation Fee:

AED 50.

Customer Refund:

AED 450.

Ticket/Commercial Policy calculates the applicable refundable amount.

Board 6 executes:

AED 450.

This prevents payment logic from independently inventing cancellation fees.

Refund of Payment Fees

Payment-related fees may have different refundability.

Examples:

- Refundable
- Non-refundable
- Proportionally refundable

The applicable policy must determine the amount before execution.

Void Architecture

Void should be used only where the provider/transaction state supports it.

Example:

Authorized → Void

not:

Settled 60 days ago → Void

In the second case:

Refund

would normally be the appropriate payment operation.

The UI should prevent invalid actions.

Reversal Architecture

Reversal is particularly important for technical failure scenarios.

Example:

Provider authorizes AED 720.

↓

TICVAI cannot complete order.

↓

System initiates:

Reversal AED 720

↓

Provider confirms.

↓

Payment:

Reversed

This prevents the customer being charged for an order that never completed.

Automatic Reversal

TICVAI may automatically initiate a reversal for predefined technical scenarios if:

- Provider supports it
- Transaction state is certain

- Policy permits it
- Idempotency is maintained

Example:

Authorization successful + order creation failed

could initiate an automatic reversal.

This is different from an agent-requested customer refund.

Refund Idempotency

Every refund operation requires its own unique reference.

Example:

REF-847291-01

If provider response times out and TICVAI retries:

the provider should receive the same idempotency/reference context where supported.

This prevents:

AED 500 refund

from accidentally becoming:

AED 1,000 refunded.

Refund Status Synchronization

Refund status may come from:

- Immediate provider response
- Webhook
- Callback
- Status inquiry
- Settlement/reconciliation

Therefore:

Submitted

must not always immediately become:

Completed.

Refund Processing Time

The UI should distinguish:

TICVAI Processing Status

Completed submission.

from:

Customer Settlement Status

Provider/bank processing.

Example:

Refund successfully submitted:

Today.

Expected customer receipt:

3–7 business days.

Where provider data supports this, the UI can show the distinction to Customer Service.

Refund Reason Codes

Configure standardized reason codes such as:

- Customer Cancellation
- Event Cancelled
- Product Unavailable
- Duplicate Payment
- Operational Error
- Pricing Error
- Technical Failure
- Service Recovery
- Fraud Resolution
- Management Exception
- Other

Reason codes should support:

- Mandatory notes
 - Approval requirements
 - Reporting
 - Finance mapping
-

Adjustment Governance

Financial adjustments should require stronger controls than normal operational edits.

Recommended states:

Draft

→ **Review**

→ **Approved**

→ **Posted**

→ **Reversed if necessary**

Never:

Edit historical transaction.

Relationship with Board 5

Board 5 records original tender allocation.

Board 6 uses it to determine refund allocation.

Example:

Board 5

AED 200 Gift Card

AED 800 Card.

Board 6

Refund AED 300.

Determines the appropriate return allocation.

Relationship with Board 2

Board 6 determines:

Refund AED 100 to original card.

Board 2/provider adapter determines:

Send refund to Provider A / Merchant Account X.

Relationship with Board 3

For card-present refunds:

Board 3 may provide terminal/device execution capability where the acquirer requires terminal-present refund handling.

Board 6 remains the **refund orchestration and governance owner**.

Relationship with Board 4

Board 4 created/processed digital payment.

Board 6 processes subsequent digital refund against the original transaction.

Relationship with Board 7

Board 6 may report:

Refund Completed — AED 500

Board 7 will verify:

Did Provider A actually deduct AED 500 from settlement?

Therefore Board 6 must retain:

- Original provider transaction
 - Refund provider transaction
 - Merchant account
 - Amount
 - Currency
 - **Status**
 - Timestamp
-

Relationship with Finance

Board 6 provides financial events such as:

- Refund
- Reversal
- Void
- Adjustment

- Stored-value restoration
- B2B credit restoration

Finance remains authoritative for:

- GL
 - Accounting periods
 - Accounts receivable
 - Settlement accounting
 - Financial reporting
-

Standard Events

Recommended events:

- REFUND_REQUESTED
 - REFUND_VALIDATED
 - REFUND_APPROVAL_REQUIRED
 - REFUND_APPROVED
 - REFUND_REJECTED
 - REFUND_SUBMITTED
 - REFUND_PENDING
 - REFUND_COMPLETED
 - REFUND_FAILED
 - REFUND_STATUS_UNKNOWN
 - REFUND_RECOVERY_STARTED
 - VOID_REQUESTED
 - VOID_COMPLETED
 - REVERSAL_STARTED
 - REVERSAL_COMPLETED
 - PAYMENT_ADJUSTMENT_CREATED
 - PAYMENT_ADJUSTMENT_APPROVED
 - PAYMENT_ADJUSTMENT_POSTED
-

Roles & Permissions

Recommended roles:

- Payment Operations
- Customer Service
- POS Supervisor

- **Venue Manager**
- **Finance Manager**
- **Commercial Manager**
- **B2B Manager**
- Payment Administrator
- System Administrator
- Auditor

Sensitive Permissions

Separately control:

- Request refund
- Approve refund
- Refund above threshold
- Change refund allocation
- Refund to alternative tender
- Execute void
- Execute reversal
- Create financial adjustment
- Approve financial adjustment
- Resolve failed refund
- Override refund restriction

Audit Requirements

Every operation must retain:

- Original Order
- Original Payment
- Refund/Reversal/Void ID
- Original amount
- Refund amount
- Tender
- Provider
- Merchant account
- Reason
- Requester
- Approver
- Policy/rule
- Override
- Provider reference
- Date/time
- Final state

The original transaction must remain immutable.

Board 6 Acceptance Criteria

Board 6 shall be considered complete when:

1. TICVAI provides a centralized Refund & Payment Adjustment Command Center.
2. Full refunds are supported.
3. Partial refunds are supported.
4. Multiple partial refunds are supported.
5. Remaining refundable balance is calculated correctly.
6. Pending refund exposure prevents over-refunding.
7. Refund requests can originate from original transactions.
8. Item-level refund selection can be supported where the source order provides it.
9. Ticket/commercial refund policy remains authoritative for refundability.
10. Payment execution rules are independently configurable.
11. Refund to original tender is supported.
12. Mixed-tender refunds are supported.
13. Original tender allocation is retained.
14. Refund allocation rules are configurable.
15. Individual tender refund limits cannot exceed original net paid amounts.
16. Stored-value restoration uses the authoritative Wallet/Gift Card service.
17. B2B credit restoration is supported.
18. Card refunds can be routed to the original provider where required.
19. Void is distinguished from refund.
20. Reversal is distinguished from refund.
21. Available financial actions depend on payment state and provider capability.
22. Technical failures can initiate governed reversal flows.
23. Refund approval thresholds are configurable.
24. Multi-level approval is supported.
25. Segregation of duties can be enforced.
26. Refund reason codes are supported.
27. Refund processing status can be monitored.
28. Provider-pending refunds are supported.
29. Failed refunds can be recovered.
30. Unknown refund status is verified before another refund is submitted.
31. Refund idempotency prevents duplicate returns.
32. Financial corrections create adjustment records rather than silently editing history.
33. Payment adjustments can require approval.
34. Complete refund decision and execution traces are retained.
35. Administrators can simulate refund outcomes before execution/configuration changes.

36. AI can identify refund anomalies and operational risks without autonomously approving sensitive refunds.
 37. Refund data is available for settlement/reconciliation.
 38. Finance receives appropriate refund/reversal/adjustment events.
 39. Every sensitive action is fully audited.
 40. All functionality respects TICVAI multi-tenancy, RBAC/PBAC, financial governance and approval requirements.
-

Board 6 — Final 10 Screens

1. **Refund & Payment Adjustment Command Center**
2. **Refund Request & Eligibility Workspace**
3. **Refund Policy & Rule Configuration**
4. **Refund Allocation & Original Tender Manager**
5. **Void, Reversal & Cancellation Manager**
6. **Refund Approval & Exception Workflow**
7. **Refund Processing, Provider Status & Recovery Center**
8. **Payment Adjustment & Financial Correction Manager**
9. **Refund Transaction Trace & Audit Investigation**
10. **Refund Simulator, Risk Analysis & AI Advisor**

Board 6 is locked as: *Refunds, Reversals, Voids & Payment Adjustments — 10 backend screens.*

Module 3 — Payment & Payment Orchestration

Board 7 — Reconciliation, Settlement & Financial Operations



Board Objective

Board 7 shall provide TICVAI with the centralized operational layer for **payment reconciliation, provider/acquirer settlement verification, transaction matching, discrepancy management, settlement tracking, fees, commissions, FX differences, payout verification, and Finance handoff.**

The key question is:

“Does what TICVAI believes was paid, refunded or reversed match what the gateway/acquirer/provider actually processed and ultimately settled?”

The responsibility split remains:

- **Board 1:** Payment method configuration
- **Board 2:** Gateway/acquirer orchestration
- **Board 3:** Card-present and terminal operations
- **Board 4:** Digital and payment-link operations
- **Board 5:** Mixed tender and B2B credit
- **Board 6:** Refunds, reversals, voids and adjustments
- **Board 7:** Reconciliation, settlement verification and financial payment operations
- **Finance Module:** Accounting, GL, financial statements, AR/AP and formal accounting

Board 7 therefore does **not replace the TICVAI Finance Module.**

It creates the trusted bridge between:

Payment Operations → Provider Settlement → Finance

Board 7 shall contain 10 backend screens.

Screen 1 — Reconciliation & Settlement Command Center

Purpose

Provide an executive and operational overview of payment reconciliation and settlement health.

KPI Cards

- TICVAI Transaction Value
- Provider Transaction Value
- Matched Transactions
- Match Rate
- Unmatched Transactions
- Unmatched Value
- Settlement Expected
- Settlement Received
- Settlement Variance
- Provider Fees
- Refund Settlement Value
- Open Exceptions

Example

TICVAI Payment Value

AED 12.84M

Matched

AED 12.71M

Unmatched

AED 130K

Match Rate

98.99%

Settlement Status

Show:

- Expected
- Pending
- Received
- Partially Received
- Variance
- Overdue

Breakdown

By:

- Provider
- Acquirer
- Merchant Account
- Venue
- Legal Entity
- Channel

- Payment Method
 - Currency
 - Settlement Date
-

Screen 2 — Reconciliation Source & Import Manager

Purpose

Manage all sources used to compare TICVAI payment records with external financial processing records.

Internal Sources

Examples:

- Payment transactions
- Refunds
- Reversals
- Voids
- Adjustments
- Mixed tender allocations
- B2B credit movements

External Sources

Support provider-specific sources such as:

- Gateway reports
- PSP settlement reports
- Acquirer settlement files
- Bank settlement files
- Provider APIs
- SFTP files
- Scheduled report imports
- Manual controlled upload

Source Configuration

For each source:

- Provider
- Merchant account
- Legal entity
- File/API type
- Frequency
- Currency
- Time zone
- Settlement calendar
- Import status
- Last successful import

Import Status

- Received

- Processing
- Completed
- Partial
- Failed
- Duplicate
- Rejected

Important

The architecture should use provider adapters rather than hard-code one reconciliation format.

Screen 3 — Transaction Matching & Reconciliation Engine

Purpose

Automatically match TICVAI payment transactions against provider/acquirer records.

Primary Match Keys

Where available:

- TICVAI Payment ID
- Provider Transaction ID
- Acquirer Reference
- Merchant Account
- Amount
- Currency
- Transaction Type
- Transaction Date

Secondary Matching

Where exact IDs are unavailable:

- Amount
- Currency
- Merchant
- Terminal
- Date/time window
- Card/payment-method metadata
- Order reference

Match Confidence

Display:

Exact Match — 100%

High Confidence — 97%

Potential Match — 82%

No Match

Match Types

- 1:1
- 1:Many

- Many:1
- Aggregated Settlement
- Partial Match
- Manual Match

Important Rule

AI/fuzzy matching can **suggest** a match.

It must not silently financially reconcile low-confidence transactions.

Screen 4 — Reconciliation Exception & Investigation Center

Purpose

Centralize all payment discrepancies requiring operational or financial investigation.

Exception Types

- Missing Provider Transaction
- Missing TICVAI Transaction
- Amount Mismatch
- Currency Mismatch
- Duplicate Transaction
- Duplicate Settlement
- Refund Mismatch
- Reversal Mismatch
- Fee Mismatch
- Settlement Missing
- Settlement Shortfall
- Unexpected Settlement
- Merchant Account Mismatch
- Timing Difference

Example

TICVAI:

AED 720

Provider:

AED 700

Variance:

AED 20

Investigation Workspace

Show:

- Payment
- Order
- Provider transaction
- Provider
- Merchant account

- Amount
- Currency
- Date
- Settlement
- Related refund/reversal
- Match result

Workflow

Detected

→ Assigned

→ Investigating

→ Resolved

→ Approved

→ Closed

Resolution Codes

Examples:

- Timing Difference
- Provider Fee
- FX Difference
- Duplicate
- Manual Adjustment
- Provider Error
- TICVAI Error
- Valid Difference

Screen 5 — Settlement & Payout Manager

Purpose

Track expected and actual settlements from payment providers/acquirers.

Settlement Record

Display:

- Settlement ID
- Provider
- Acquirer
- Merchant Account
- Legal Entity
- Settlement Period
- Gross Sales
- Refunds
- Chargebacks
- Fees
- Adjustments

- Reserve/Holdback
- Net Expected
- Net Received
- Currency
- Settlement Date

Example

Gross Sales:

AED 1,000,000

Refunds:

—AED 50,000

Provider Fees:

—AED 22,000

Chargebacks:

—AED 5,000

Expected Settlement

AED 923,000

Actual:

AED 923,000

Status**Reconciled****Settlement States**

- Expected
- Scheduled
- Processing
- Received
- Partially Received
- Overdue
- Variance
- Reconciled
- Closed

Screen 6 — Fees, Commission, FX & Settlement Economics**Purpose**

Provide visibility into the financial deductions and differences between gross payment value and net settlement.

Fee Types

Support:

- Fixed transaction fee
- Percentage fee
- Authorization fee

- Capture fee
- Refund fee
- Chargeback fee
- Cross-border fee
- Wallet/APM fee
- Terminal fee where relevant
- Other provider fee

Example

Payment:

AED 1,000

Provider Fee:

2.5%

Fixed Fee:

AED 1

Expected Fee

AED 26

Net

AED 974

Expected vs Actual

Compare:

- Contracted fee
- Calculated expected fee
- Provider charged fee
- Variance

FX

Where currencies differ, show:

- Selling currency
- Payment currency
- Provider transaction currency
- Settlement currency
- Exchange rate
- FX fee
- Settlement conversion

Important Boundary

Board 7 explains payment economics.

The Finance Module remains authoritative for formal accounting treatment.

Screen 7 — Merchant Account & Settlement Calendar Manager**Purpose**

Manage operational settlement expectations across providers, merchant accounts and legal entities.

Hierarchy

Tenant

↓

Legal Entity

↓

Provider / Acquirer

↓

Merchant Account

↓

Settlement Configuration

Configuration

For each merchant account:

- Merchant Account ID
- Provider
- Acquirer
- Legal Entity
- Country
- Currency
- Bank destination reference
- Settlement frequency
- Settlement delay
- Cut-off time
- Weekend handling
- Holiday handling
- Reserve policy
- **Status**

Example

Merchant:

Water Park Dubai

Settlement:

T+2

Cut-off:

23:00 GST

Weekend:

Next Business Day

Calendar

Display:

- Transaction period
- Expected settlement date

- Actual settlement date

- **Status**

This allows TICVAI to distinguish:

not yet due

from:

overdue settlement.

Screen 8 — Settlement Posting, Finance Handoff & Close Manager

Purpose

Control when reconciled payment information becomes ready for downstream financial posting and period close.

Operational Flow

Payments

↓

Provider Records

↓

Matched

↓

Exceptions Resolved

↓

Settlement Verified

↓

Ready for Finance

Finance Handoff Package

Include:

- Legal entity
- Settlement period
- Payment totals
- Refund totals
- Reversal totals
- Chargeback totals
- Provider fees
- FX differences
- Adjustments
- Net settlement
- Merchant account
- Currency

Status

- Not Ready

- Reconciliation Pending
- Exception Pending
- Ready for Finance
- Exported/Posted
- Finance Accepted
- Rejected
- Closed

Close Controls

Prevent settlement period closure when:

- **Critical unmatched transactions remain**
- Unapproved adjustments exist
- Settlement variance exceeds tolerance

unless authorized override is used.

Screen 9 — Reconciliation Audit, Trace & Evidence Center

Purpose

Provide a complete audit trail explaining how any payment became reconciled and financially closed.

Search

By:

- Payment ID
- Order
- Settlement
- Provider Transaction
- Refund
- Merchant Account
- Reconciliation Batch
- Adjustment
- Date

Example Trace

PAY-847291

TICVAI Payment:

AED 720

↓

Provider Transaction:

AED 720

↓

Exact Match:

✓

↓

Settlement Batch:

SET-2026-0987

↓

Provider Fee:

AED 18

↓

Expected Net:

AED 702

↓

Bank Settlement:

AED 702

↓

Reconciled

Evidence

Store/reference:

- Import batch
- Provider report
- API response
- Match rule
- Manual match
- Exception resolution
- Approval
- Adjustment
- Finance handoff

Audit

Capture:

- User
- Action
- Previous state
- New state
- Reason
- Timestamp
- Approval

Screen 10 — Reconciliation Simulator, Forecast & AI Operations Advisor

Purpose

Provide simulation, forecasting and AI-assisted analysis across reconciliation and settlement operations.

Simulation Inputs

Select:

- Provider

- Acquirer
- Merchant account
- Date range
- Currency
- Transaction volume
- Payment value
- Refund value
- Chargeback assumptions
- Contracted fees
- Settlement cycle

Example

Gross Payments:

AED 5,000,000

Refunds:

AED 200,000

Expected Fees:

AED 115,000

Chargebacks:

AED 20,000

Expected Net Settlement

AED 4,665,000

Expected Settlement Date

07 September

Settlement Forecasting

Forecast:

- Expected settlement
- Expected cash receipt
- Provider fees
- Refund deductions
- Chargeback deductions
- Currency conversion
- Merchant account payout

Example:

Date	Expected
Sep 03	AED 1.2M
Sep 04	AED 980K
Sep 05	AED 1.4M
Sep 06	AED 760K

This can support Treasury/Finance cash visibility without duplicating Finance forecasting.

AI Reconciliation Advisor

AI can identify:

Provider A has 46 transactions totaling AED 81,250 with no matching settlement record beyond the normal T+2 window.

Merchant Account M-204 was charged an average effective fee of 3.1%, compared with the configured contractual rate of 2.6%.

92 unmatched transactions appear to belong to Settlement Batch SET-20481 with 98.7% matching confidence.

Refund settlement delays for Provider B increased significantly this week.

AI Actions

AI may:

- Suggest transaction matches
- Identify anomalies
- Detect settlement delays
- Detect abnormal fees
- Identify duplicate records
- Forecast settlement
- Recommend investigation
- Explain reconciliation variance

AI should **not autonomously write off material differences or approve financial adjustments.**

Core Reconciliation Architecture

The recommended model is:

TICVAI Operational Ledger

Payment

Refund

Void

Reversal

Adjustment

↓

Reconciliation Engine



Provider / Acquirer Records

↓

Settlement Engine



Settlement / Bank Evidence

↓

Finance Handoff

This gives TICVAI three distinct levels of confirmation:

1. Payment processed

2. Provider confirms transaction

3. Money settled

These should not be treated as the same event.

Three-Level Financial State

This is a critical design principle for Development team.

A payment can be:

Operationally Successful

Customer paid.

but:

Not Yet Reconciled

Provider file has not arrived.

and:

Not Yet Settled

Funds have not reached the expected settlement.

Therefore do not use one generic:

Payment = Complete

to represent all three layers.

Recommended State Model

Transaction State

Authorized / Captured / Refunded / etc.

Reconciliation State

- Not Due
- Awaiting Provider Data
- Matched
- Partially Matched
- Unmatched
- Exception
- Reconciled

Settlement State

- Not Expected Yet
- Expected
- Processing
- Received
- Partial
- Variance
- Overdue
- Settled

These states must remain independently visible.

Reconciliation Batches

Board 7 should support reconciliation batches.

Example:

Batch

REC-20260902-STRIPe-UAE-01

Contains:

- Provider
- Merchant account
- Period
- Currency
- Imported records
- TICVAI records
- Matched
- Unmatched
- Exceptions
- Total variance

Batch Lifecycle

Created

→ **Imported**

→ **Matching**

→ **Exception Review**

→ **Approved**

→ **Reconciled**

→ **Closed**

This creates a strong operational control point.

Matching Tolerances

Exact matching should be preferred.

However, controlled tolerances may be necessary.

Example:

Timestamp difference:

±5 minutes.

FX rounding:

AED 0.01.

Settlement aggregation:

Allowed according to provider format.

Tolerance rules must be configurable and audited.

Timing Differences

Not every mismatch is an error.

Example:

TICVAI captures payment:

31 August 23:58.

Provider settlement cut-off:

23:00.

Transaction appears in:

1 September settlement batch.

The reconciliation engine should understand configured cut-off rules rather than incorrectly flagging the transaction as missing.

Provider File Duplication

If the same settlement file is uploaded twice, TICVAI must detect:

- File ID
- Hash/fingerprint
- Provider
- Settlement period
- Existing batch

and prevent duplicate financial processing.

Provider API Duplication

The same applies to APIs.

Provider events should have:

- Provider Event ID
- Provider Transaction ID
- Idempotency/deduplication handling

to prevent duplicated reconciliation records.

Gross vs Net Settlement

Providers may settle:

Gross

Sales paid separately, fees invoiced separately.

or:

Net

Sales minus:

- Refunds
- Fees
- Chargebacks
- Reserves

Board 7 must support both settlement models.

Aggregated Settlement

One bank deposit may represent thousands of transactions.

Example:

Bank Settlement:

AED 923,000

contains:

2,817 sales

– 73 refunds

– fees

– 4 chargebacks.

Therefore TICVAI needs:

Transaction-level reconciliation

plus:

Settlement-level reconciliation.

These are related but different.

Settlement Variance

Recommended calculation:

Actual Settlement

–

Expected Settlement

=

Settlement Variance

Example:

Expected:

AED 923,000.

Received:

AED 918,000.

Variance:

–AED 5,000

Status:

Investigation Required

Settlement Tolerance

Tenant can configure:

Absolute tolerance:

AED 5.

and/or:

Percentage tolerance:

0.01%.

Above tolerance:

Exception

Within tolerance:

May be automatically classified according to approved policy.

Provider Fees

The system should compare:

Expected Fee

based on configured commercial agreement

against:

Actual Fee

reported by provider.

This creates an important commercial control.

Example:

Expected:

AED 18,200.

Actual:

AED 21,600.

Difference:

AED 3,400

This should be visible rather than silently accepted.

Board 2 vs Board 7 Provider Economics

There is a deliberate distinction.

Board 2

Uses provider costs for:

routing decisions.

Example:

Provider A cheaper than Provider B.

Board 7

Uses actual provider costs for:

financial verification.

Example:

Did Provider A charge the contracted amount?

Therefore these screens are complementary, not duplicates.

Multi-Currency Reconciliation

Board 7 should distinguish:

- Order Currency
- Payment Currency
- Provider Currency

- Settlement Currency
- Reporting Currency

Example:

Customer pays:

USD 100

Provider processes:

USD 100

Merchant settlement:

AED 365.80

Finance reporting:

AED

The FX rate and conversion fees should remain traceable.

Refund Reconciliation

Board 6 says:

Refund:

AED 500 — Completed

Board 7 verifies:

Provider settlement:

–AED 500

If not:

Refund Settlement Exception

This prevents operational success from being confused with financial completion.

Reversal Reconciliation

Similarly:

Board 6:

Reversal Completed

Board 7:

Confirm provider/acquirer actually reflected the reversal.

Chargebacks

Board 7 should ingest chargeback settlement impacts because they affect settlement.

However, **Board 8** will own:

- Chargeback case management
- Fraud
- Dispute lifecycle
- Risk analytics

Board 7 only reconciles the financial effect.

B2B Credit

B2B credit does not necessarily generate provider settlement.

Board 5 records:

B2B Credit AED 10,000

Finance receives:

Accounts Receivable exposure

Board 7 must distinguish this from gateway-settled tender and avoid expecting an acquirer settlement.

Cash Payments

Similarly:

POS cash:

AED 500

does not appear in a card provider settlement.

Cash reconciliation may integrate with Finance/POS cash management.

Board 7 should classify the tender correctly instead of generating a provider mismatch.

Finance Handoff Architecture

Recommended:

Board 7 produces a governed financial settlement package.

Example:

Settlement Package

Legal Entity:

TICVAI UAE Client A

Provider:

Provider A

Period:

01–02 Sep

Sales:

AED 1,000,000

Refunds:

–AED 50,000

Chargebacks:

–AED 5,000

Fees:

–AED 22,000

Net:

AED 923,000

Status:

Reconciled

↓

Finance consumes this package for accounting.

Period Close

Payment Operations should have a controlled close process.

Example

Settlement period:

01 September.

Open exceptions:

3

Variance:

AED 80

Tolerance:

AED 5

Result:

Cannot Close

unless:

Authorized Override

with reason and approval.

Reopening

If a previously closed reconciliation period must be reopened:

Require:

- Permission
- Reason
- Approval
- **Audit**
- Version/history

Do not silently modify a closed settlement period.

Evidence Retention

Board 7 should retain or reference reconciliation evidence according to client retention policy.

Examples:

- Provider settlement report
 - Import batch
 - Bank evidence
 - API response reference
 - Adjustment approval
 - Reconciliation result
-

Relationship with Finance Module

This distinction should remain explicit in the design.

Board 7 owns

- Payment reconciliation
- Provider reconciliation
- Settlement matching
- Fee verification
- Payment exceptions
- Settlement readiness
- Payment financial trace

Finance owns

- General Ledger
- Accounting journals
- Financial periods
- Accounts receivable/payable
- Balance sheet
- P&L
- Tax accounting
- Financial statements

Board 7 should therefore integrate with Finance rather than recreate it.

Standard Events

Recommended events include:

- RECONCILIATION_BATCH_CREATED
 - PROVIDER_DATA_IMPORTED
 - RECONCILIATION_STARTED
 - TRANSACTION_MATCHED
 - TRANSACTION_UNMATCHED
 - MATCH_SUGGESTED
 - RECONCILIATION_EXCEPTION_CREATED
 - RECONCILIATION_EXCEPTION_RESOLVED
 - SETTLEMENT_EXPECTED
 - SETTLEMENT_RECEIVED
 - SETTLEMENT_VARIANCE_DETECTED
 - SETTLEMENT_RECONCILED
 - FINANCE_HANDOFF_READY
 - FINANCE_HANDOFF_COMPLETED
 - RECONCILIATION_PERIOD_CLOSED
 - RECONCILIATION_PERIOD_REOPENED
-

Roles & Permissions

Recommended roles:

- Payment Operations
- Reconciliation Analyst
- Finance Operations
- **Finance Manager**
- Treasury
- Payment Administrator
- **Revenue Manager**
- System Administrator
- Auditor

Sensitive Permissions

Separately control:

- Import provider settlement data
- Manually match transaction
- Unmatch transaction
- Resolve exception
- Approve reconciliation
- Create settlement adjustment
- Approve settlement variance
- Close reconciliation period
- Reopen period
- Send Finance handoff
- Override settlement tolerance

Audit Requirements

Every reconciliation action should capture:

- Reconciliation Batch
- Payment
- Provider Transaction
- Settlement
- Original state
- New state
- Match method
- Match confidence
- Variance
- User/system
- Reason
- Approval
- Date/time

Manual matches and overrides must be especially visible.

Board 7 Acceptance Criteria

Board 7 shall be considered complete when:

1. TICVAI provides a centralized Reconciliation & Settlement Command Center.
2. Internal payment records can be reconciled against provider/acquirer data.
3. Provider APIs can be used as reconciliation sources where available.
4. Provider settlement files can be imported.
5. Controlled manual imports are supported.
6. Duplicate import files are detected.
7. Provider events are deduplicated.
8. Reconciliation batches are supported.
9. Exact transaction matching is supported.
10. Controlled secondary/fuzzy matching is supported.
11. Match confidence is visible.
12. Low-confidence AI matches require human review.
13. 1:1 and aggregated matching scenarios are supported.
14. Unmatched transactions create exceptions.
15. Amount mismatches are identified.
16. Currency mismatches are identified.
17. Refund mismatches are identified.
18. Reversal mismatches are identified.
19. Duplicate settlements are identified.
20. Timing/cut-off differences can be recognized.
21. Expected settlements are calculated.
22. Actual settlements can be recorded/imported.
23. Partial settlements are supported.
24. Settlement variance is calculated.
25. Settlement tolerances are configurable.
26. Overdue settlements can be detected.
27. Gross and net settlement models are supported.
28. Expected provider fees can be calculated.
29. Actual provider fees can be compared.
30. Fee variance is visible.
31. Multi-currency reconciliation is supported.
32. FX differences remain traceable.
33. Merchant-account settlement calendars are configurable.
34. Settlement periods can be closed.
- 35. Critical exceptions can block close.**

36. Controlled close overrides are audited.
 37. Reconciled settlement packages can be handed to Finance.
 38. Complete reconciliation evidence and transaction trace are retained.
 39. AI can suggest matches, anomalies and settlement forecasts but cannot autonomously write off material financial differences.
 40. All functionality respects TICVAI multi-tenancy, legal-entity separation, RBAC/PBAC, approvals and financial audit requirements.
-

Board 7 — Final 10 Screens

1. **Reconciliation & Settlement Command Center**
2. **Reconciliation Source & Import Manager**
3. **Transaction Matching & Reconciliation Engine**
4. **Reconciliation Exception & Investigation Center**
5. **Settlement & Payout Manager**
6. **Fees, Commission, FX & Settlement Economics**
7. **Merchant Account & Settlement Calendar Manager**
8. **Settlement Posting, Finance Handoff & Close Manager**
9. **Reconciliation Audit, Trace & Evidence Center**
10. **Reconciliation Simulator, Forecast & AI Operations Advisor**

Board 7 is locked as: *Reconciliation, Settlement & Financial Operations — 10 backend screens.*

Module 3 — Payment & Payment Orchestration

Board 8 — Payment Risk, Fraud, Chargebacks, Disputes, Analytics & AI



Board Objective

Board 8 is the **final governance, protection, intelligence and optimization layer** of the TICVAI Payment & Payment Orchestration module.

It shall provide centralized capabilities for:

- Payment fraud monitoring
- Real-time payment risk assessment
- Transaction risk rules
- Velocity and behavioral controls
- Suspicious payment detection
- Chargeback management
- Payment disputes
- Evidence and representation workflows
- Risk lists and payment controls
- Fraud and chargeback analytics
- Provider/payment performance analytics
- AI-assisted anomaly detection
- AI risk recommendations
- Payment executive intelligence

The key question is:

“How does TICVAI protect payment transactions from fraud and abuse, manage disputes and chargebacks, understand payment performance, and continuously improve payment operations using governed AI?”

The responsibility split is now complete:

Board 1 — What payment methods are available?

Board 2 — Which provider/acquirer processes the transaction?

Board 3 — How are physical card/terminal payments processed?

Board 4 — How are digital and remote payments processed?

Board 5 — How are multiple tenders and B2B credit combined?

Board 6 — How are refunds, reversals and corrections handled?

Board 7 — Did provider settlement financially reconcile?

Board 8 — **Is payment behavior safe, where are the risks, how are disputes handled, and how is the entire payment operation performing?**

Board 8 shall contain 10 backend screens.

Screen 1 — Payment Risk & Fraud Command Center

Purpose

Provide real-time operational visibility over fraud, suspicious transactions, payment risk and chargebacks across the TICVAI ecosystem.

KPI Cards

- Payment Attempts
- Approved Transactions
- Declined Transactions
- Risk-Blocked Transactions
- Risk-Review Transactions
- Suspected Fraud Value
- Confirmed Fraud Value
- Chargebacks
- Chargeback Value
- Chargeback Rate
- Open Disputes
- High-Risk Transactions

Risk Distribution

Show:

- Low Risk
- Medium Risk
- High Risk
- **Critical Risk**

Fraud Trend

Analyze:

- Fraud attempts
- Blocked fraud
- Confirmed fraud
- Chargebacks
- False positives

Breakdown

By:

- Venue
- Channel
- Payment method
- Provider
- Country
- Customer segment
- Product
- Device
- Merchant account

Real-Time Alerts

Example:

17 payment attempts from the same payment instrument across 9 customer accounts within 12 minutes.
 Chargeback rate for B2C Channel increased above the configured threshold.

Screen 2 — Payment Risk Rule & Decision Engine**Purpose**

Allow administrators to configure deterministic payment risk rules and actions.

Rule Inputs

Rules may evaluate:

- Transaction amount
- Currency
- Channel
- Venue
- Product
- Payment method
- Provider
- Customer/account
- Transaction history
- Failed attempts
- Refund history
- Chargeback history
- Device context
- Location/country context
- Time
- Velocity
- Risk signals received from payment providers

Example

IF

Failed Payment Attempts > 5

within:

10 minutes

AND

Same Payment Instrument

THEN

Action:

Require Additional Verification

Actions

Support:

- Allow
- **Monitor**
- Require additional verification
- Route to manual review
- Block
- Apply velocity restriction
- Trigger alert

Rule Priority

Rules should support:

- Priority
- Effective date
- Tenant
- Channel
- Scope
- Override
- **Status**

Screen 3 — Velocity, Behavioral & Transaction Risk Controls

Purpose

Manage high-frequency and abnormal payment activity.

Velocity Dimensions

Examples:

Per Payment Instrument

Maximum:

5 transactions / 10 minutes.

Per Customer

Maximum:

10 failed attempts / hour.

Per Device

Maximum:

20 transactions / hour.

Per Account

Maximum:

AED 25,000 / day.

Dimensions

Support:

- Customer
- Account
- Device
- Payment instrument reference/token
- IP/network signals where legitimately available
- Email/account reference
- Phone/account reference
- Venue
- POS
- Terminal
- Operator
- Merchant account

Behavioral Controls

Identify patterns such as:

- Repeated small payments
- Rapid retries
- Multiple accounts using the same instrument
- Unusual transaction amount
- Unusual purchasing velocity
- Excessive refunds
- Repeated payment failures
- Sudden channel change

Important

The system should evaluate context and avoid treating every unusual transaction as fraud.

Screen 4 — Risk Lists, Signals & Payment Control Center

Purpose

Provide centralized management of approved payment risk lists and signals.

List Types

Support controlled lists such as:

- Trusted customer/account
- Trusted device
- Blocked customer/account

- Blocked instrument reference
- High-risk device
- High-risk network indicator
- High-risk merchant/terminal/operator
- Review list

Avoid Terminology Ambiguity

Rather than relying only on “whitelist/blacklist,” TICVAI should use:

- **Trusted**
- **Blocked**
- **Review Required**

Record

For each entry:

- List type
- Identifier/reference
- Reason
- Source
- Created by
- Effective date
- Expiry
- **Status**
- Approval
- Related case

Provider Signals

Ingest risk signals from:

- PSP
- Gateway
- Acquirer
- Fraud provider
- 3DS/authentication provider

while maintaining TICVAI's own decision trace.

Screen 5 — Fraud Alert, Investigation & Case Management

Purpose

Convert suspicious activity into controlled investigation cases.

Fraud Case

Display:

- Case ID
- Risk level
- Customer
- Transaction

- Order
- Payment method
- Provider
- Amount
- Currency
- Device
- Channel
- Risk score
- Rules triggered
- AI signals
- Related transactions

Case Workflow**Detected**

→ **Alert Created**

→ **Assigned**

→ **Investigating**

→ **Decision**

→ **Resolved**

→ **Closed**

Investigation Actions

Authorized users may:

- Add note
- Link transactions
- Link customers/accounts
- Request additional review
- Mark confirmed fraud
- Mark legitimate
- Add risk-list entry
- Escalate
- Close case

Related Activity

Show:

- Previous transactions
 - Failed payments
 - Refunds
 - Chargebacks
 - Devices
 - Payment instruments
 - Customer accounts
-

Screen 6 — Chargeback & Dispute Command Center

Purpose

Provide centralized operational management for provider/acquirer chargebacks and payment disputes.

KPI Cards

- New Chargebacks
- Open Chargebacks
- Chargeback Value
- Chargeback Rate
- Response Due
- Won
- Lost
- Accepted
- Represented
- Recovery Value

Chargeback Record

Display:

- Chargeback ID
- Original Payment
- Order
- Customer
- Provider
- Acquirer
- Merchant account
- Amount
- Currency
- Reason code
- Received date
- Response deadline
- **Status**

States

- Received
- Under Review
- Evidence Required
- Evidence Ready
- Submitted
- Representment
- Won
- Lost
- Accepted
- Closed

Important

Board 7 reconciles the **financial deduction**.

Board 8 manages the **actual dispute/chargeback case**.

Screen 7 — Chargeback Evidence & Representment Workspace**Purpose**

Help Payment Operations collect, review and submit the evidence needed to respond to disputes.

This is particularly important for TICVAI because ticketing transactions generate strong operational evidence.

Potential Evidence

Depending on dispute type:

- Order confirmation
- Payment record
- Ticket
- Ticket issuance timestamp
- Ticket terms
- Customer acceptance
- Refund policy
- Access-control scan
- Entry timestamp
- Gate/location
- Seat/reservation
- Event attendance
- Membership usage
- Customer communication
- Delivery confirmation
- Digital-ticket activation
- Device/transaction references

Example

Customer disputes:

“Service not received.”

TICVAI can show:

Ticket:

TKT-847291

Access Scan:

Valid Entry

Gate:

Main Entrance

Date:

12 August 2026

Time:

14:32

Evidence Package**Payment**

-

Order

-

Ticket

-

Access Scan

-

Terms

↓

Representment Package**Governance**

Evidence should be reviewed before submission where required.

Screen 8 — Payment Performance & Conversion Analytics**Purpose**

Provide comprehensive analytics across the complete Payment & Payment Orchestration module.

This screen is not only fraud analytics.

It provides **payment-business performance intelligence**.

Core Metrics

- Payment Attempts
- Authorization Rate
- Capture Rate
- Success Rate
- Decline Rate
- Failure Rate
- Abandonment Rate
- Refund Rate
- Chargeback Rate
- Fraud Rate
- Average Transaction Value
- Processing Time

Analyze By

- Provider
- Acquirer
- Payment method
- Card scheme
- Digital wallet

- Venue
- Channel
- Merchant account
- Currency
- Country/market
- Device/terminal type

Payment Funnel

Checkout

100%



Payment Method Selected

92%



Payment Started

88%



Authentication

82%



Authorized

79%



Captured

78%

Provider Comparison

Example:

Provider	Auth Rate	Avg Time	Cost	Chargeback
Provider A	96.8%	1.2s	2.4%	0.28%
Provider B	94.1%	1.8s	2.1%	0.35%
Provider C	97.2%	1.1s	2.7%	0.19%

This information can feed Board 2 routing optimization.

Screen 9 — AI Fraud, Anomaly & Payment Intelligence Center

Purpose

Provide governed AI/ML capabilities for fraud detection and payment anomaly identification.

AI Signals

AI can evaluate patterns such as:

- Transaction velocity

- Amount deviation
- Device/account relationships
- Failed-payment patterns
- Refund behavior
- Chargeback history
- Channel behavior
- Provider anomalies
- Merchant anomalies
- Transaction timing
- Customer behavior

Example AI Insight

Transaction PAY-849201 has an elevated risk profile because the payment instrument was used across six new accounts in 18 minutes.

Another Example

Provider B's authorization rate declined from 96.1% to 88.4% during the last 45 minutes without a corresponding increase in customer risk scores. This may indicate a provider issue rather than fraud.

This distinction is extremely valuable.

AI Classification

Potential outcomes:

- Normal
- **Monitor**
- Elevated Risk
- Investigation Recommended
- Critical

Explainability

Every AI recommendation should explain:

- Main contributing signals
- Confidence
- Historical comparison
- Recommended action

Screen 10 — Payment Executive Intelligence, Risk Simulator & AI Advisor**Purpose**

Provide executive-level intelligence and a safe simulation environment for payment, fraud and risk strategies.

Executive Scorecard

Show:

- Total Payment Value
- Payment Success Rate
- Authorization Rate
- Payment Cost

- Refund Rate
- Fraud Rate
- Chargeback Rate
- Settlement Accuracy
- Provider Performance
- Payment Conversion

Example

Payment Value:

AED 48.2M

Success Rate:

94.8%

Fraud Rate:

0.21%

Chargeback Rate:

0.34%

Payment Cost:

2.42%

Settlement Match:

99.2%

Risk Simulation

Administrators should be able to test rule changes before publishing.

Example

Current rule:

Block after 5 failed attempts / 10 minutes.

Proposed:

Block after 3 failed attempts / 10 minutes.

Simulation Result

Fraud blocked:

+AED 42K

Legitimate payments blocked:

+AED 118K

False-positive increase:

+2.7%

Recommendation

Do not activate globally.

Test on high-risk segment first.

This is precisely where AI should help the business rather than simply produce a fraud score.

AI Payment Advisor

The AI advisor can provide recommendations across the complete module.

Examples:

Provider A has the highest authorization rate for Mobile App wallet transactions but currently receives only 30% of eligible traffic.

Provider B experienced an abnormal decline spike between 14:00–14:45. Consider temporary routing adjustments subject to Board 2 controls.

Refund rate for Product Group X increased 37% following the latest schedule configuration.

Chargebacks tagged “service not received” have a 78% representment success rate when Access Control evidence is attached.

Fraud Rule FR-102 blocked AED 18K of suspected fraud but also declined an estimated AED 61K of legitimate sales.

Critical Architecture — Risk Decision Point

Payment risk must be evaluated at appropriate stages.

Recommended flow:

Customer / POS / Channel

↓

Payment Request

↓

TICVAI Payment Risk Evaluation

↓

Decision:

Allow

or

Additional Verification

or

Review

or

Block

↓

Payment Orchestration

↓

Provider / Acquirer

↓

Provider Risk / Authentication

↓

Payment Result

↓

TICVAI Post-Transaction Monitoring

This means risk is not only a reporting process after payment.

Pre-Transaction vs Post-Transaction Risk

TICVAI should distinguish:

Pre-Transaction

Used to make an immediate payment decision.

Examples:

- Velocity
- Blocked instrument
- Transaction amount
- Customer risk
- Device risk

Post-Transaction

Used for investigation and learning.

Examples:

- Chargeback
- Refund behavior
- Linked-account behavior
- Fraud confirmation
- Settlement anomaly

Both feed the risk intelligence layer.

Risk Decision Architecture

Recommended decision order:

1. Hard Security Controls

Example:

Blocked instrument.

↓

2. Deterministic Risk Rules

Example:

Velocity exceeded.

↓

3. Provider Signals

Example:

Provider high-risk signal.

↓

4. AI/ML Risk Assessment

↓

5. Business Policy

↓

Final Decision

This keeps AI from bypassing hard controls.

Risk Score

Where used, the score should have a governed meaning.

Example:

0-29

Low.

30-59

Medium.

60-79

High.

80-100

Critical.

However, each client should be able to configure thresholds and actions.

A score by itself should never be the only information shown to an operator.

False Positive Management

This is essential.

A fraud system that blocks too much legitimate revenue is not successful.

Board 8 must track:

Fraud Prevention

versus

Legitimate Revenue Lost

Example:

Rule:

High transaction velocity.

Fraud blocked:

AED 50K.

Legitimate sales incorrectly blocked:

AED 130K.

Result:

Rule may require optimization.

Risk Rule Shadow Mode

I strongly recommend supporting:

Shadow Mode

A new rule can evaluate live transactions without actually blocking them.

Example:

Rule would have:

Blocked:

382 transactions.

Estimated fraud:

AED 42K.

Potential legitimate loss:

AED 118K.

Then the client decides whether to activate it.

This is especially valuable for AI-generated rules.

AI Rule Lifecycle

Recommended:

AI Detects Pattern

↓

Recommendation

↓

Draft Rule

↓

Simulation

↓

Shadow Mode

↓

Human Approval

↓

Limited Rollout

↓

Monitor

↓

Full Activation

This should be the standard TICVAI AI governance model.

Payment Risk & Access Control Integration

TICVAI has an advantage because it also controls ticket validation.

Example:

Customer disputes:

"I never attended."

But TICVAI has:

- Ticket activation
- QR validation
- Gate
- Timestamp
- Venue

- Access event

This information can materially strengthen dispute evidence.

Boundary

Access Control remains authoritative for access events.

Board 8 consumes those events as evidence.

CRM Integration

CRM may provide:

- Customer tenure
- Customer segment
- Account history
- Loyalty status
- Previous purchase behavior

Board 8 may use these as approved risk signals.

It should not duplicate CRM customer profiles.

Device Intelligence

Where privacy and technical architecture permit, Board 8 may use device-level signals such as:

- Device reference
- Known/new device
- Repeated device usage
- Multiple accounts from device
- Transaction frequency

Sensitive fingerprinting approaches should be governed according to applicable privacy requirements and client policy.

Provider Risk vs TICVAI Risk

Payment providers may already provide fraud/risk products.

TICVAI should not unnecessarily recreate every provider capability.

Instead:

Provider Risk

provides:

- Provider risk score
- Authentication
- Network intelligence
- Card/payment intelligence

TICVAI Risk

adds:

- Ticketing context
 - Venue context
 - Customer behavior
-

- Access behavior
- Membership
- Product behavior
- Cross-provider history

This makes TICVAI's risk layer particularly useful.

Chargeback Lifecycle

Recommended:

Chargeback Received

↓

Case Created

↓

Reason Classified

↓

Evidence Required

↓

Evidence Collected

↓

Decision:

Accept

or

Represent

↓

Provider Submission

↓

Won / Lost

↓

Financial result sent to:

Board 7 Reconciliation

Chargeback Deadlines

Every chargeback must include:

- Received date
- Response deadline
- Days remaining
- SLA status

Alerts:

Due Soon

Due Today

Overdue

A dispute should not be lost simply because the response deadline was missed.

Chargeback Evidence Automation

Where evidence is available internally, TICVAI should automatically **suggest** it.

Example:

Chargeback:

Service Not Received

AI/system identifies:

- ✓ Ticket issued
- ✓ Ticket activated
- ✓ Gate scan recorded
- ✓ Customer entered venue
- ✓ Confirmation email sent

The system can assemble a draft evidence package.

Human review can approve submission.

Chargeback Reason Analytics

Analyze chargebacks by:

- Fraud
- Service not received
- Duplicate processing
- Refund not received
- Product/service issue
- Authorization issue
- Other provider reason codes

This helps TICVAI distinguish:

Payment fraud

from:

Operational/service problem.

Payment Performance Analytics Architecture

Board 8 should combine operational metrics from Boards 1–7.

Board 1

Method availability.

Board 2

Provider routing.

Board 3

Terminal performance.

Board 4

Digital conversion.

Board 5

Mixed-tender behavior.

Board 6

Refunds.

Board 7

Settlement.

↓

Board 8 — Payment Intelligence

This provides the executive view across the entire module.

Metric Governance

Metrics must have standardized definitions.

Example:

Authorization Rate

Authorized Transactions / Authorization Attempts

Payment Success Rate

Successfully Completed Payments / Valid Payment Attempts

Refund Rate

Should specify whether measured by:

- Transaction count
- Payment value

Chargeback Rate

Should specify:

- Count basis
- Value basis
- Provider/acquirer calculation basis

This avoids different dashboards reporting different values for the same KPI.

AI Architecture Recommendation

For Board 8, use a combination of:

Rules Engine

For deterministic controls.

Statistical/ML Models

For:

- Fraud scoring
- Anomaly detection
- Pattern recognition
- Propensity/risk classification

LLM Layer

For:

- Explaining fraud cases

- Summarizing investigations
- Drafting evidence narratives
- Natural-language analytics
- Explaining anomalies
- Suggesting configuration changes

The LLM should **not be the primary real-time fraud-scoring engine**.

Real-Time Performance Requirement

Risk evaluation used in payment authorization must be low latency.

The architecture should therefore support:

Payment Request

→ Risk Decision

→ Payment Orchestration

without materially degrading checkout performance.

Heavy AI analysis can operate asynchronously after transaction completion where appropriate.

Fraud Feedback Loop

The system should learn from outcomes.

Example:

Risk Decision:

Medium.

↓

Payment Approved.

↓

30 days later:

Chargeback confirmed fraud.

↓

Outcome:

Confirmed Fraud

↓

Risk analytics/model feedback.

Similarly:

High risk.

↓

Manual Review.

↓

Confirmed legitimate.

↓

Outcome:

False Positive

This feedback is critical for improving the system.

Standard Events

Recommended events include:

- PAYMENT_RISK_EVALUATED
 - PAYMENT_RISK_ALLOWED
 - PAYMENT_RISK_REVIEW_REQUIRED
 - PAYMENT_RISK_BLOCKED
 - RISK_RULE_TRIGGERED
 - VELOCITY_THRESHOLD_EXCEEDED
 - FRAUD_ALERT_CREATED
 - FRAUD_CASE_OPENED
 - FRAUD_CONFIRMED
 - TRANSACTION_CONFIRMED_LEGITIMATE
 - CHARGEBACK_RECEIVED
 - CHARGEBACK_EVIDENCE_REQUIRED
 - CHARGEBACK_EVIDENCE_SUBMITTED
 - CHARGEBACK_WON
 - CHARGEBACK_LOST
 - RISK_RULE_SIMULATED
 - RISK_RULE_SHADOW_MODE_STARTED
 - AI_RISK_RECOMMENDATION_CREATED
-

Roles & Permissions

Recommended roles:

- Payment Risk Analyst
- Fraud Analyst
- Chargeback Analyst
- Payment Operations
- Finance Operations
- **Finance Manager**
- Payment Administrator
- Security Administrator
- Compliance/Audit
- System Administrator

Sensitive Permissions

Separately control:

- Create risk rule
- Activate risk rule

- Block payment instrument
 - Add trusted reference
 - Override risk decision
 - Mark confirmed fraud
 - Accept chargeback
 - Submit representment
 - Change dispute evidence
 - Activate AI-recommended rule
 - Change fraud threshold
-

Audit Requirements

Every risk-sensitive decision should retain:

- Payment ID
- Customer/account reference
- Risk score
- Rules triggered
- Provider signals
- AI/model result
- Final decision
- Override
- User/system
- Reason
- Timestamp
- Model/rule version where applicable

This provides explainability even months after the original transaction.

Board 8 Acceptance Criteria

Board 8 shall be considered complete when:

1. TICVAI provides a centralized Payment Risk & Fraud Command Center.
2. Real-time payment risk rules are supported.
3. Risk rules can evaluate transaction, customer, channel and payment context.
4. Velocity controls are configurable.
5. Behavioral risk indicators are supported.
6. Trusted, blocked and review lists are supported.
7. Provider fraud/risk signals can be consumed.
8. TICVAI risk decisions remain independently traceable.
9. Allow, verify, review and block decisions are supported.
10. Fraud alerts can generate investigation cases.
11. Related payments/customers/devices can be linked during investigation.

12. Fraud cases have controlled workflows.
13. Confirmed fraud and confirmed legitimate outcomes are captured.
14. False-positive measurement is supported.
15. Risk rules can operate in simulation mode.
16. Risk rules can operate in shadow mode.
17. AI-generated risk recommendations require governed activation.
18. Chargebacks can be imported/received from providers.
19. Chargeback cases are centrally managed.
20. Chargeback reason codes are retained.
21. Response deadlines are tracked.
22. Evidence requirements are supported.
23. TICVAI order and ticket evidence can be associated with disputes.
24. Access Control scan evidence can be associated where relevant.
25. Evidence packages can be assembled.
26. Representment workflow is supported.
27. Chargeback win/loss outcomes are recorded.
28. Chargeback financial impacts integrate with Board 7 reconciliation.
29. Payment authorization analytics are available.
30. Payment conversion analytics are available.
31. Provider performance can be compared.
32. Payment-method performance can be compared.
33. Refund, fraud and chargeback analytics are available.
34. Metrics use governed definitions.
35. AI/ML can identify payment anomalies and risk patterns.
36. AI explanations include contributing signals and confidence.
37. AI does not bypass deterministic hard security controls.
38. Risk decisions and model/rule versions remain auditable.
39. Executive payment intelligence covers the complete Boards 1–7 payment lifecycle.
40. All functionality respects TICVAI multi-tenancy, privacy controls, RBAC/PBAC, approvals, audit requirements and financial governance.

Board 8 — Final 10 Screens

1. **Payment Risk & Fraud Command Center**
2. **Payment Risk Rule & Decision Engine**
3. **Velocity, Behavioral & Transaction Risk Controls**
4. **Risk Lists, Signals & Payment Control Center**
5. **Fraud Alert, Investigation & Case Management**
6. **Chargeback & Dispute Command Center**
7. **Chargeback Evidence & Representment Workspace**
8. **Payment Performance & Conversion Analytics**
9. **AI Fraud, Anomaly & Payment Intelligence Center**

10. Payment Executive Intelligence, Risk Simulator & AI Advisor

Board 8 is locked as: *Payment Risk, Fraud, Chargebacks, Disputes, Analytics & AI* — 10 backend screens.

With Board 8, the **Payment & Payment Orchestration Module** is structurally complete: 8 boards × 10 screens = approximately 80 backend screens.

